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The Effect of Digital Quality on Customer Satisfaction and Brand Loyalty Under Environmental Uncertainty: Evidence from the Banking Industry

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Abstract: This study investigates whether digital quality significantly shapes customers' perceived service quality and examines the relative importance of its components—system, information, and service quality—as brand equity factors. Specifically, it explores the impact of digital quality on customer satisfaction and brand loyalty and analyzes how its individual elements interact with environmental uncertainty. A survey was conducted from February to March 2023 via Google Forms, targeting 406 Korean bank users. Quota sampling was used to ensure balanced age distribution, and after removing insincere responses, 330 valid samples were analyzed using structural equation modeling with Smart-PLS. The findings reveal three main insights. First, among perceived quality dimensions, brand image, customer orientation, and physical quality significantly influenced customer satisfaction and, in turn, enhance brand loyalty. In contrast, brand awareness and brand availability were not significant determinants. Second, digital quality—comprising system, information, and service quality—indirectly boosted brand loyalty by enhancing customer satisfaction. Third, while environmental uncertainty positively affected customer satisfaction, it did not significantly moderate the effects of digital quality components. These results imply that in the digital banking era, customer valuation is increasingly influenced by digital service channel quality and brand image rather than traditional brand equity elements like company size. Nonetheless, the continued relevance of physical quality underscores the importance of managing both physical and digital service environments. Therefore, multi-channel service quality management is essential for banks aiming to strengthen brand equity and maintain customer loyalty in uncertain environments.

Keywords: digital quality; environmental uncertainty; customer satisfaction; brand loyalty; banking industry

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1. Introduction

The keywords representing the changes in the contemporary management environment surrounding companies are the acceleration of technological change and the diversity and personalization of customer needs. Rapid technological changes have also affected the market, shortening product life cycles, while intensifying price competition and expanding opportunities for imitation are diluting the meaning of cost leadership strategy and product differentiation strategy, which are the original strategies of companies. In this environment, 'brand' can be cited as a source that can provide companies with a competitive advantage.

The core value of a brand becomes even more important in the case of services characterized by intangibility and invisibility. In particular, in the case of the financial service industry such as banking, new financial services are emerging along with changes in technology and demand following the advent of the fourth industrial revolution era, and as non-financial companies based on fintech enter the financial service market, financial service companies' inter-Korean cooperation and competition are intensifying [1]. Against this, financial institutions are trying to find a successful answer in brand equity as one of their main means of competition, and are stimulating fast digital transformation of financial institutions to improve brand equity [2,3].

In accordance with such changes in the business environment of financial institutions, customers' sensitivity to digital service quality ('perceived quality') as a component of brand equity recognized by customers, is increasing in addition to traditionally emphasized components such as trust-based brand availability, customer orientation, and physical environment. However, if you look at previous academic studies on service quality, there are many studies on the components of traditional 'perceived quality' but few studies on 'digital quality', which is being emphasized at the same time. In this regard, this study aimed at exploring the effect of 'digital quality' on customers' 'perceived quality' and its value as a component of brand equity by adding 'digital quality' to the components of 'perceived quality' dealt with in previous studies on the banking industry.

To this end, a survey was conducted with banking customers, and an empirical analysis using statistics was conducted to identify the impact and path of perceived quality on customer satisfaction and brand loyalty as well as the bank's brand equity based on the overall research results. The purpose of this study is to provide theoretical and practical implications for the establishment of related strategies. In addition, digital quality is subdivided into system quality, digital information quality, and service quality, and the differences in the relationship between customer satisfaction and customer loyalty of the detailed components of digital quality are examined.

In accordance with the above research purpose, this study consists of a total of six sections. Section 1 describes the problem statement and research purpose as an introduction, and Section 2 derives the core research hypotheses that this paper intends to deal with from the theoretical background and review of previous studies. Section 3 describes the study design, including the research model and variable measurement, and Section 4 summarizes the results of the statistical analysis. In Section 5, the theoretical and practical implications of the research results are discussed, and in Section 6, the limitations of this study and future research themes are presented as a conclusion.

2. Theoretical Background and Hypothesis Generation

2.1. Theoretical Background

2.1.1. Brand Equity

Brand equity can be defined from either a financial or a customer-based perspective. In the former, it is conceptualized as the brand's market value reflected on the balance sheet—that is, the brand value—or the additional value attributed to the brand [4]. In the latter, brand equity refers to the various effects that brand knowledge—including brand awareness, associations (images), and beliefs—has on consumers' responses to the brand [5,6]. It can also be defined in terms of consumer attachment to the brand, often referred to as brand strength or brand loyalty [7], as well as brand response, which includes brand judgments and brand feelings [8].

Brand equity consists of brand awareness and brand image in the cognitive stage, brand preference and response in the attitude stage, and brand loyalty in the behavioral stage. These stages are understood to have a sequential precedence relationship in the

order of cognition, attitude, and behavior. Brand power [9] is formed by adding brand availability (functional power), including economic quality and 'service quality', such as relationship quality and physical quality.

2.1.2. Service Quality

Research on service quality begins with the study of Grönroos [10], and this model is defined as a subjective criterion rather than an objective criterion, that is, the meaning of 'service quality perceived' by consumers [11]. Perceived quality refers to the quality level of a specific brand recognized through consumers' direct and indirect experiences and acquired information, and the concept of perceived quality varies according to scholars. Parasuraman et al. [7] defines it as a customer's judgment of the excellence and superiority of a service or product, which is a type of subjective attitude that is different from actual and objective quality, and according to Zeithaml [8], it is the overall excellence or superiority of a product or service. It is defined as the customer's evaluation of the brand, and a perceived high quality means that customers recognize the differentiation and superiority of the brand through long-term experience related to the brand. For consumers, perceived quality is more important than objective quality when making a service purchasing decision.

Service quality is a 'perceived quality' based on the subjectivity of consumers' experiences [10] and corresponds to brand judgments, a component of brand response. Unlike brand feelings, which are an emotional response based on socially accepted notions (warmth, fun, security, pride, sense of achievement, etc.) about the brand, brand judgment is a consumer's personal opinion based on the brand's performance and image association, including perceived quality, reliability, and intention to purchase or use [8]. Positive brand judgment and brand sentiment reinforce the value of brand equity.

Service quality is conceptually divided into technical quality, which is result quality, and functional quality, which is process quality [10]; in practice, these are modeled using the SERVQUAL model [12] or modified quality models [13–16]. In this article, the SERVPERF model [17] is used to examine the relationship between service quality determinants, customer satisfaction, and customer loyalty based on the logic that service quality should be measured as service performance based on performance among modified quality models.

Traditionally, in the banking industry, interaction quality and physical environment quality are important due to offline face-to-face transactions with customers. Interaction quality can be defined as the quality of interpersonal relationships between customers and service employees [18], and as a quality related to the service delivery process, it consists of items related to the attitudes and behaviors of employees. Specifically, as a result of comprehensively reviewing previous studies, it can be measured by items such as 'kindness', 'reliability', 'professionalism', and 'responsiveness (quickness)' of bank tellers. The physical environment quality is defined as "the service environment that is the background of service delivery" [18] and includes the quality of physical facilities and the appearance of buildings. Specifically, it can be measured by items such as 'comfort of the surrounding environment', 'state-of-the-art facilities', 'interior design', and 'employees' appearance'.

Similar to interaction quality and physical environment quality, the service quality that customers pay attention to in the banking industry is result quality. Result quality is a concept similar to 'service product', 'actual service', or 'technical quality', and can be defined as a "service product that remains as a result after the service delivery process is over" [18]. Specifically, result quality can be measured by items such as 'stability of financial products', 'benefits of financial products', 'interest rates', and 'commissions'. Regardless of the quality of other service aspects, the nature of the financial industry makes customers highly sensitive to result quality [19,20]. In the financial industry, result

quality, particularly in terms of financial benefits, is often the most critical factor influencing customer satisfaction and retention. Even if the service delivery process is friendly and professional, customers are likely to switch to another financial institution if the benefits of financial products—such as interest rates or fee structures—are perceived as inferior.

In addition, there is also the quality of service resulting from the characteristics of the financial industry. Typically, the degree to which a customer can use a particular bank, the customer's image of a bank branch, and the ease and convenience of using a branch are important factors in measuring the customers' perceived quality of a bank. According to actual research, even if the result quality or interaction quality is excellent, if there is inconvenience in use, bank customers are likely to change their business partners. For example, if the main bank branch is far away, customers are likely to be motivated to use the nearby branch of another bank. In practice, most bank users maintain transactions with two or more different banks.

Furthermore, due to the nature of the financial industry, where credit rating is important, bank awareness, bank size, and reliability of the bank brand are emphasized. These are important factors for measuring the perceived quality of banks. On the other hand, with the advent of the digital age; the development of IT technology, fintech, and big tech; and the promotion of digitalization, the convenience of digitalized financial transactions that guarantee security and safety is emerging as a new component of service quality. The significance of digital quality as a component of service quality is increasing.

2.1.3. Digital Transformation

Digital transformation is a broad concept within digital innovation that drives change across society, the economy, and corporate business through the application of IT technologies [21]. At the corporate level, it also refers to a narrower concept—a continuous process of adapting to disruptive changes in the business environment by leveraging digital capabilities to develop, produce, and deliver new products and services.

In addition, at the corporate level, passive conceptualization [22] views the behavior of applying technology, such as building software and systems, to secure competitive advantages, improve profitability [22], and enhance the network of organizational members, including their customer experience. Active conceptualization [23] is a digital technology integration process that seeks to fundamentally change the business level.

With digital transformation, companies can expect corporate growth that enhances the core value of the brand through consumer-oriented big data analysis, and fashion companies can increase consumer satisfaction through digital experiences that reflect consumer experiences. In addition, real-time communication between consumers and producers is possible, forming a market where demand and supply meet in a new way, providing customized products and services and helping to establish personalized product strategies as the boundaries between online and offline collapse [24]. In other words, key changes brought about by digital transformation—such as super-intelligence, hyper-realism, and hyper-connectivity—are reshaping purchasing channels and methods. These advancements optimize information sharing, enhance operational efficiency for companies, and strengthen communication between businesses and consumers, ultimately helping to reduce market failures. Ultimately, the expected effects of digital transformation include improvements in business performance [25,26] as well as changes in the production environment, consumption environment, distribution environment, and service environment [27,28].

Due to the development and combination of fintech and IT technology, online channels such as the Internet and mobile transactions are becoming more important in the banking industry as well as traditional offline channels in transactions with customers. Also, the most important thing is to provide hyper-personalized and customized services to

customers due to the development of digital technology. This digitalization of the banking industry provides customers with a new type of service quality called 'digital quality'. Digital quality here refers to the evaluation of the quality of reactions, emotions, and actions that customers feel in the process of accessing an online platform provided by a company using digital devices (Internet, mobile, chatbot, etc.) based on information and communication technology.

Hwang Jae-young, Lee Eung-bong, and Kim Jong-hwan [29], who analyzed the quality of library services due to digitalization, conducted a preceding study by organizing digital service quality into three independent variables: system quality, information quality, and service quality. It was revealed that system quality consisted of accessibility, security, and ease of use; information quality consisted of sufficiency, diversity, up-to-dateness, accuracy, and usefulness; and service quality consisted of customer support, customization, and reliability. In this study, building on the research of Hwang Jae-young, Lee Eung-bong, and Kim Jong-hwan [29], the changes in product and service quality resulting from digital transformation promoted by companies are categorized into system quality, information quality, and service quality. The purpose of this study is to examine the relationship between customers' perceived digital quality and their value judgments regarding overall service quality.

2.1.4. Uncertainty

The concept of uncertainty is often interpreted similarly to volatility or risk; however, while volatility and risk refer to actual phenomena occurring at a specific point in time, uncertainty pertains to the recognition of the possibility of such phenomena occurring [30,31]. Liesch, Welch, and Buckley [32], in their examination of uncertainty and risk through the lens of internalization theory in the context of outward foreign direct investment, distinguish the two by stating that uncertainty requires acclimatization, whereas risk requires accommodation. Furthermore, Nicolaou and McKnight [33] classified perceived uncertainty from an organizational perspective as a resultant factor that arises from the level of trust in the counterpart and the perceived risk. Similarly, Luo [34], in his strategic analysis of China's industrial environment, conceptualized the changing environment itself as volatile and explained that volatility implies uncertainty insofar as it inherently includes unpredictable elements.

Uncertainty, as reflected in its multilayered conceptualization, takes many forms. In the field of business administration, it has been categorized into various types, including environmental uncertainty [35,36], technological uncertainty [37–40], state uncertainty [41], effect uncertainty [41], response uncertainty [41], competitive uncertainty [42], supplier uncertainty [27], information uncertainty [33], structural uncertainty [34], and firm-specific uncertainty [43].

Despite the long-term research on uncertainty, academic understanding of the function of uncertainty is still conflicting [38,39]. Williamson [44] and Alvarez and Barney [45], who explored uncertainty from the transaction cost perspective, explained that when uncertainty is recognized, compensation for investment is not properly obtained due to the manifestation of opportunistic psychology in economic actors. Similarly, from the perspective of situational theory, Song and Montoya-Weiss [38] classified uncertainty as a negative factor on the grounds that it prevents a company's strategic behavior from being directed to performance in a situation where technological uncertainty is highly perceived. In contrast, in terms of corporate competitive advantage, Hartman [46], Abel [47], Caballero [48], and Teece et al. [49] argued that entrepreneurs make efforts to secure a new competitive advantage in order to respond to the changing business environment. The positive aspects of uncertainty were emphasized. In a similar context, Luo [34] also

described an uncertain environment as an environment in which opportunities are yet to be manifested.

2.2. Literature Review and Hypotheses

2.2.1. Perceived Quality and Customer Satisfaction

Although perceived service quality and customer satisfaction are distinct concepts, numerous previous studies have identified perceived quality as an antecedent variable that is closely linked to customer satisfaction [50,51]. The two concepts are strongly interconnected. Customer satisfaction, which is associated with various outcomes pursued by companies, is a core concept in marketing and has received significant attention from both academia and practice [52,53].

According to the accumulated research results, the majority opinion is that service quality is an antecedent variable of customer satisfaction [50] and that the relationship between service quality and customer satisfaction is generally positive [54]. The antecedent relationship of service quality to customer satisfaction is clearly shown in the SERVPERF model based on performance rather than the difference in service quality [17,55]. For example, Dabholkar et al. [56] paid attention to the mediating role of customer satisfaction between service quality and customer loyalty. They proved the statistical significance of the model by depicting how customer satisfaction fully mediates the relationship between service quality and resultant behavioral intentions according to the formation of customer loyalty. In addition, many studies have shown that customer satisfaction increases when their perceived service quality increases, which is linked to repurchase intention and word of mouth, which improves loyalty and corporate performance [56–58].

Karatepe et al. [59] stated that the quality of banking services is closely related to customer satisfaction. Yuhanju and Song Kwang-seok [60] also empirically found that the components of service quality in the banking industry have a positive effect on customer satisfaction. On the other hand, Kim Ji-young [61] categorized banking service quality into multiple channels and empirically analyzed the effects of quality evaluation across these channels on customer satisfaction and loyalty, based on customer age. Excluding telebanking service quality, the study found that high perceived quality in Internet and mobile banking, along with the physical environment and human services, had a positive impact on both customer satisfaction and loyalty.

Customer satisfaction has been recognized as a central concept in bank marketing strategies in relation to financial and non-financial performance pursued by banks, and the antecedent relationship of service quality to customer satisfaction is clearly shown in the SERVPERF model [55].

This study defines service quality as brand awareness, brand availability, brand image, customer orientation, and physical environment quality. Among these, brand awareness represents the brand equity that allows a firm to differentiate itself from others, thereby inducing customers to favor a particular brand [4,5,55,62]. The measurement of brand awareness is divided into recall, which is a state in which one can recall without providing a clue, and recognition, which means whether there is an image associated with or concordance with memory for the provided brand cue [55].

The brand expansion effect through such brand awareness forms brand loyalty. Efforts to build the value of a bank's brand equity begin with strengthening brand awareness, and banks with larger assets tend to have higher brand awareness in the market. Brand awareness based on the size of a bank's assets, the so-called 'bullet-to-fail', has traditionally been the basis for customer trust. Therefore, based on previous studies and market attributes, this article established the first basic hypothesis that a positive (+) relationship will be established between brand awareness, brand image, and customer satisfaction as follows.

Hypothesis 1-1. *The customer's perceived quality (brand awareness) of banking services will be positively associated with customer satisfaction.*

Hypothesis 1-2. *The customer's perceived quality (brand image) of banking services will be positively associated with customer satisfaction.*

2.2.2. Brand Availability

The third component of perceived quality is brand availability. Brand availability can be viewed in the same context as distribution. In other words, if a consumer wants to use a particular brand, he or she must use a store that has that brand. Having a wide distribution network increases the possibility that consumers can use a particular brand. Expansion of the distribution network means expanding the points where consumers can come in contact with the brand. Aaker [63] found that the distribution network affects brand awareness and association, and high brand awareness means that the company's business area and distribution network are wide, and members of the distribution network try to treat with a brand well known to customers as perceived quality increases. In other words, high brand availability indicates that consumers prefer the brand due to its high perceived quality. On the other hand, Yoo et al. [64] studied the effect of distribution network among marketing mix factors on perceived quality and brand loyalty. In their study, the expansion of the distribution network was found to have a positive effect on brand equity.

Meanwhile, as banks belong to the financial industry, customers consider interest rates or risks when deciding to entrust their financial assets. Therefore, the outcome quality provided by a bank is one of the main qualities perceived by customers.

Hypothesis 2. *The customer's perceived quality (brand availability) of banking services will be positively associated with customer satisfaction.*

2.2.3. Customer Orientation

The fourth component of perceived quality is customer orientation. Customer orientation is one of relationship quality factors and is defined as attitudes and behaviors aimed at satisfying customer needs at the level of employee interaction with customers [65,66]. Customer orientation is described as the philosophy and behavior of identifying and understanding the needs of the target customer and adapting the responses of the sales organization to meet the needs of the customer better than its competitors do, which is to satisfy the customer's needs better than its competitors. To create a competitive advantage [67], customer orientation is recognized as a key factor in the success of service organizations. Service companies build relationships with customers by listening to them, paying attention to their needs, providing accurate and relevant information, and keeping their promises [17].

Swan, Trawick, and Silva [68] viewed customer orientation in four dimensions, and the more the service provider understands the customer and delivers good service to the customer, the more the customer evaluates the service quality. An important factor in achieving customer satisfaction is to increase the customer orientation of salespeople. Employees with a high level of customer orientation engage in behaviors that increase customer satisfaction, and these behaviors benefit both firms and consumers [65,69]. A salesperson must know the customer's needs well, and when these needs are met, the customer can be satisfied. Service providers with high customer orientation show behaviors to increase customer satisfaction, and customer-oriented behaviors are important because they lead to the establishment of long-term relationships between service providers and customers [65]. In the same context, Ha Hong-yeol and Choi Chang-bok [70] demonstrated

that customer orientation is very important for service companies, and customer satisfaction and loyalty can be improved by increasing customer orientation in banking services.

In the banking industry, due to the nature of the service, salespersons who come into contact with customers interact with customers and affect the management and control of customers' service experience [71] as well as customers' overall evaluation of the service [72]. In addition, from the point of view of a financial product salesperson, a salesperson's behavior can be one of the important factors that can differentiate the company from competitors [73]. Therefore, customer orientation has a significant impact on a bank's profitability through customer satisfaction [74] and is accepted as an essential factor for competitive advantage [75]. Based on these studies, this article established the following hypotheses.

Hypothesis 3. *The customer's perceived quality (customer orientation) of banking services will be positively associated with customer satisfaction.*

2.2.4. Physical Environment Quality

The physical environment quality, measured by 'comfort of the surrounding environment', 'state-of-the-art facilities', and 'interior design', also forms the customers' perceived quality [76]. Since services have a different evaluation process or purchasing decision-making method than products, marketing managers of service companies must implement marketing strategies different from those of companies that produce products [77,78]. There are many factors that affect a consumer's purchase decision when purchasing or using a particular service. One of them is the physical environment. If a service company does not manage this tangible evidence, customers will have a false perception of the service, resulting in an unfavorable perception, and the company's marketing strategy will not be able to achieve its purpose [79]. Service marketers should consider the effects of these physical environments on customer satisfaction, employees, and customer service [80,81]. Bitner [81] defined the service physical environment as an objective and physical factor that a company can control, expressing it as the service scape. As customers use services within the store, the role of the physical environment is to provide customers with informational cues about service quality and product assortment. As the importance of the environment within service stores is being highlighted and the influence on customer satisfaction is increasing, it is necessary to manage it more efficiently [79]. The physical environment is a quality factor that affects customer evaluation [18], and Joo and Kwon [82] saw that the physical environment quality of a bank has a significant effect on customer satisfaction.

It should be noted that in the banking industry, both the provision of services through direct contact with customers at branches and the fintech-based online service path are open at the same time, so challenges to the effect of individual components of service quality on customer satisfaction are expected to take place. Hence, the following hypothesis is proposed:

Hypothesis 4. *The customer's perceived quality (physical quality) of banking services will be positively associated with customer satisfaction.*

2.2.5. Digital Quality

The fifth component of perceived quality is digital quality. Recently, the financial industry—including the banking sector—has been facing declining interest income, stricter regulations, rising security risks, increasing customer demands, and higher costs due to the growth of non-face-to-face services. Additionally, competition with non-financial industries and fintech startups is intensifying [83]. Therefore, managing digital quality

from the customer's perspective, in line with a bank's digital transformation capabilities based on digital technologies, has become increasingly important.

In the same context, most of the previous studies have concluded that digital quality improvement following digital transformation of banks enhances customer satisfaction and value. Representatively, Kim Woo-jin [27] said that in order to induce a positive impact in the customer-centered experience process and increase customer satisfaction, it is necessary to provide the best experience to customers through securing the quality of provided digital content and convenience. Seck and Philippe [84] saw that the quality of physical and online routes affects customer satisfaction for banking services. Kwak Ki-young and Lee Yu-jin [85] identified perceived usefulness and immediate connectivity as key factors influencing the perception of mobile banking service quality. Kim [61] revealed that customers' perceived quality of a bank's multi-channel service affects customer satisfaction.

The conceptualization of digital quality in this study draws upon the DeLone and McLean Information Systems Success Model [86], which identifies system quality, information quality, and service quality as critical dimensions influencing user satisfaction. These three dimensions have been widely validated in digital service contexts, including online banking, where they affect customer perceptions of usability, trustworthiness, and responsiveness. System quality refers to the performance and technical functionality of an information system. It includes characteristics such as ease of use, reliability, flexibility, response time, and accessibility [86]. It evaluates how well the system itself operates and supports the user experience. Information quality represents the quality of the outputs produced by the system—i.e., the information [86]. It includes accuracy, timeliness, completeness, relevance, and consistency. It reflects how useful and trustworthy the system's information is for decision making. Service quality reflects the support provided to users by the IS department or service provider [86]. It involves responsiveness, reliability, empathy, and technical competence. It assesses how well users are supported during and after system use.

Based on the preceding studies, this article is expected to enhance customer satisfaction by satisfying customer needs in all components of digital quality, such as system quality, information quality, and service quality. In this study, system quality refers to the usability, reliability, and overall performance of digital platforms, which directly affect the ease of customer interaction. Information quality involves the accuracy, timeliness, and relevance of the information provided, helping customers make informed decisions. Service quality includes responsiveness, personalization, and support in digital interactions, contributing to a more satisfying customer experience. System quality—including usability, reliability, and speed—is crucial for digital banking because it directly affects how efficiently customers can perform transactions. For instance, a mobile banking app that takes too long to load or frequently crashes can lead to customer frustration, decreasing satisfaction regardless of other service factors [87,88]. Information quality is another critical component. Customers rely on accurate and timely information to make informed financial decisions. Inconsistent balance updates or outdated loan interest rate information can lead to confusion or even financial losses, undermining trust in the bank [89,90]. Service quality in the digital context refers to prompt, personalized, and helpful responses from customer support channels such as live chat or AI chatbots. A delay in resolving urgent issues—like failed wire transfers—may escalate dissatisfaction [91,92]. Thus, high-quality digital platforms not only support functional interaction but also shape emotional customer experience, influencing satisfaction [93,94]. These considerations form the basis for the following hypotheses.

Hypothesis 5-1. *The customer's perceived quality (digital system quality) of banking services will be positively associated with customer satisfaction.*

Hypothesis 5-2. *The customer's perceived quality (digital information quality) of banking services will be positively associated with customer satisfaction.*

Hypothesis 5-3. The *customer's perceived quality (digital service quality) of banking services will be positively associated with customer satisfaction.*

2.2.6. Customer Satisfaction and Brand Loyalty

Customer satisfaction is an emotion evoked by a response based on customer experience and interaction experience related to a service [95,96]. Due to its linkage with corporate performance, it has long received attention from the academic and practical worlds. Many studies on the customer satisfaction–customer loyalty relationship [56,57,97] suggest that when the customers' perceived quality improves, customer satisfaction increases, leading to customer loyalty. For instance, Ragunathan and Irwin [98] claim that as customer satisfaction increases, brand loyalty increases. In addition, high customer satisfaction improves the loyalty of existing customers and leads to the creation of new customers according to the word of mouth of satisfied existing customers. The positive relationship between customer satisfaction and customer loyalty is also accompanied by ripple effects such as reduced price sensitivity, reduced marketing failure costs, reduced new customer creation costs, and improved corporate reputation [99].

Bowen and Chen [100] proposed three approaches to understanding customer loyalty: the behavioral approach, the attitudinal approach, and the integrated approach. The behavioral approach views loyalty as a tendency for repeat purchases, while the attitudinal approach focuses on customer preferences or psychological commitment. The integrated approach combines both perspectives. Meanwhile, Oliver [101] presented a four-stage model of customer loyalty, consisting of cognitive loyalty, affective (emotional) loyalty, conative (intentional) loyalty, and behavioral loyalty.

Customer loyalty encompasses factors related to word of mouth, in addition to purchase intention. For bank customers, satisfaction increases when perceived service quality improves, which in turn is linked not only to (re)purchase intention but also to word-of-mouth intention—both of which enhance customer loyalty and corporate performance [56–58,97]. Satisfied customers are more likely to not only continue purchasing or make additional purchases from the bank but also to engage in word-of-mouth behavior by sharing information about the bank's service quality with others [57].

The positive relationship between customer service quality and loyalty was confirmed in both a study by Wong and Sohal [102] on retail chain stores and a study by Koh Hwajeong and Jeong Soon-hee [50] on the quality of financial services and customer loyalty. A study by Dabholkar, Hepherd, and Thorpe [56], which paid attention to the mediating role of customer satisfaction between service quality and customer loyalty, proved a model that mediates the relationship between customer satisfaction and the resultant behavioral intention of forming customer loyalty. In this paper, we adopted Oliver's model [103], which is considered to be the most comprehensive evaluation model for the relationship between customer satisfaction and loyalty, as it complementarily accepts the word-of-mouth effect perspective to evaluate customer loyalty. It consists of repurchase intention (action) and word of mouth intention (action). Based on this understanding, this article anticipates that customer satisfaction with banking services is likely to lead to brand loyalty, which is an essential antecedent for building brand equity [104]. Therefore, the following hypotheses about the relationship between customer satisfaction and loyalty were tested.

Hypothesis 6. *The customer's satisfaction with the banking services will be positively associated with brand loyalty.*

2.2.7. Environmental Dynamism (Uncertainty)

The substantive direct and moderating effects of environmental uncertainty vary across studies [105,106]. The contingent role of environmental dynamics shown in Schilke's [107] analysis exerts a linear moderating effect on corporate competitive advantages. Changes in technology and consumer needs increase uncertainty in the banking industry. Accordingly, banks try to simultaneously pursue strategic exploitation and exploration to survive in the market in the short term and to acquire sustainable competitive advantages in the long term [108]. Environmental uncertainty has a positive function, strengthening competitiveness by promoting the digital transformation of banks, while digital quality reinforcement has a positive effect on consumers' perceived quality, resulting in a moderating effect. Thus, two hypotheses are established as follows:

Hypothesis 7-1. *Uncertainty will be positively associated with customer satisfaction.*

Hypothesis 7-2. *Uncertainty will moderate the linear relationship between digital quality and customer satisfaction.*

3. Research Methodology

3.1. The Unit of Analysis and Measurements of Variables

The unit of analysis in this study was not a banking organization but consumers who use banking services. Individual-level surveys were conducted. The questionnaire drew upon the literature surrounding brand equity, customer satisfaction, brand loyalty, and digital transformation in the banking industry. Construct measures were adopted from previous research by Aaker, Keller, Kwahk, and Lee; Moghavvemi, Lee, and Lee; Seck and Philippe; Parasuraman, Zeithaml, and Berry; and von Leipzig et al. [8,16,84,85,109–113]. Every construct consisted of three questions, and the entire questionnaire included eight constructs—brand awareness, relationship quality, physical quality, economic quality, digital quality, customer satisfaction, brand loyalty, and uncertainty—and 58 questions, excluding demographic questions. The questionnaire used a 7-point Likert scale, which has been proven to be superior to other types of scales [114].

3.2. Research Model

The purpose of this study is to empirically examine the relationship in which brand equity components of the banking industry affect brand loyalty through customer satisfaction. Based on the theoretical background and research hypotheses, the elements of brand equity were composed of brand awareness, brand availability, brand image, customer orientation, physical environment quality, and digital quality. To investigate whether these six components affect customer satisfaction and further brand loyalty through customer satisfaction, and to examine whether uncertainty moderates the relationship between digital quality and customer satisfaction, the following research model was designed (see Figure 1).

3.3. Operational Definition of Variables and Measures

The conceptual definition and measurement variables of each variable were as follows, and all measurement items used in the survey employed a 7-point Likert scale from 1 to 7, representing 'strongly disagree' to 'strongly agree' (see Table 1).

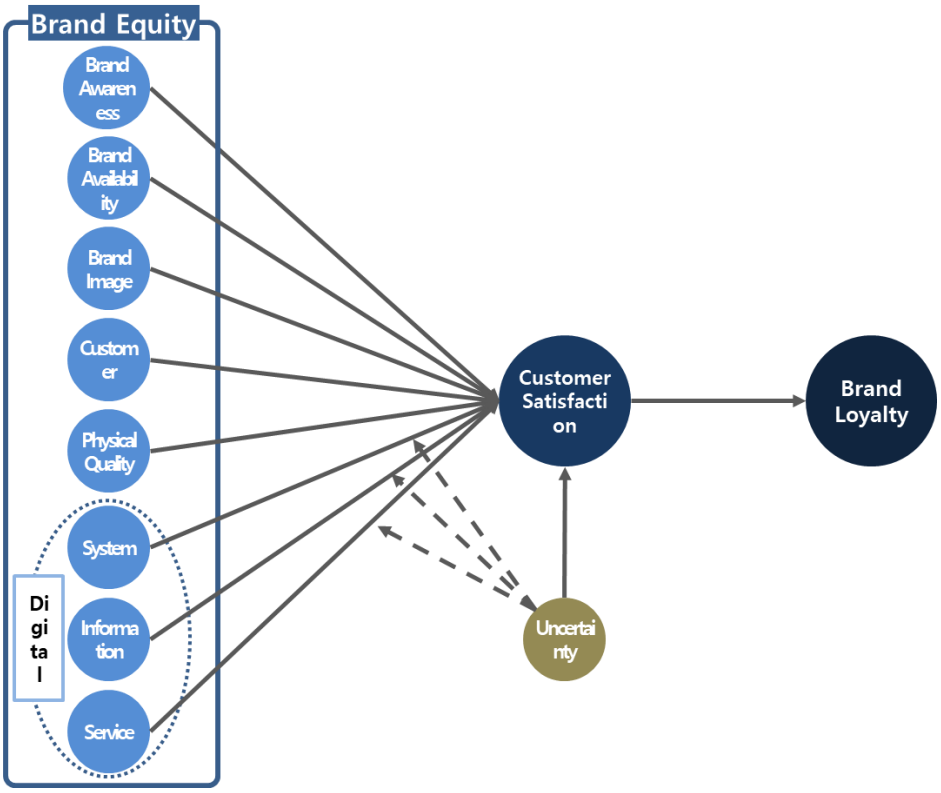


Figure 1. Research model. Note: A line indicates a direct effect, while a dotted line refers to a moderating effect. Source: Created by the author.

Table 1. Conceptual definition and measure of brand equity.

Factor	Conceptual Definition	Measure
Brand Awareness	The degree to which consumers are aware of or recall a particular brand	I can often come across advertisements and promotions of my main bank. My main bank's brand is easy to remember. My main bank's logo is easily distinguishable. My main bank is a representative brand in the banking industry. I am well informed of the characteristics of my main bank.
Brand Availability	The degree to which a customer can use a particular bank	My main bank has more branches compared to competing brands. My main bank offers more diverse financial products than other banks. My main bank provides more advantageous (profitable) financial products than other banks. My main bank's products are less risky than other banks.
Brand Image	Overall impression customers have of a particular bank	My main bank has a sophisticated image. My main bank gives me a sense of familiarity. My main bank has a trustworthy image. My main bank gives an image leading the times. My main bank is reputable.
Customer Orientation	The extent to which customers are satisfied with understanding and satisfying their needs	The staff at my main bank's branches is friendly. The staff at my main bank's branches is very responsive to questions. The staff at my main bank's branches wants to know what customers want. The staff at my main bank's branches offers good solutions to customers. The staff at my main bank's branches takes care of my problems quickly.
Physical Environmental Quality	Objective/physical factors that companies can control	The facilities of my main bank's branches are clean. My main bank's branches are a space where I want to stay for a long time. My main bank's branches are generally convenient to use. My main bank's branches are generally aesthetically pleasing.

Source: Created by the author.

3.3.1. Brand Awareness

Brand awareness is defined as 'the extent to which consumers recognize or recall a specific brand', and the five measurement items are based on the questionnaire by Aaker [115]: 'I can often come across advertisements and promotions of my main bank', 'my main bank's brand is easy to remember', 'my main bank's logo is easily distinguishable', 'my main bank is a representative brand in the banking industry', and 'I am well informed of the characteristics of my main bank' (see Table 1).

3.3.2. Brand Availability

Brand availability is defined as 'the extent to which a customer can use a particular bank', and based on the survey items of Smith [116] and Yoo et al. [64], it was evaluated with four items: 'my main bank has more branches compared to competing brands', 'my main bank offers more diverse financial products than other banks', 'my main bank provides more advantageous (profitable) financial products than other banks', 'my main bank's products are less risky than other banks' (see Table 1).

3.3.3. Brand Image

Brand image is defined as 'the overall impression customers have of a particular bank', and based on Cho and Lim [117], it was evaluated with five items: 'my main bank has a sophisticated image', 'my main bank gives me a sense of familiarity', 'my main bank has a trustworthy image', 'my main bank gives an image leading the times', and 'my main bank is reputable'.

3.3.4. Customer Orientation

Customer orientation is defined as 'the degree of customer satisfaction with understanding and satisfying customer needs', and based on the research of Saxe and Weitz [118] and Williams [119], it was evaluated with five items: 'the staff at my main bank's branches is friendly', 'the staff at my main bank's branches is very responsive to questions', 'the staff at my main bank's branches wants to know what customers want', 'the staff at my main bank's branches offers good solutions to customers', and 'the staff at my main bank's branches takes care of my problems quickly'.

3.3.5. Physical Environment Quality

The physical environment is defined as 'objective/physical factors that can be controlled by the company', and based on the research of Bitner [81]; Baker, Grewal, and Parasuraman [120]; and Lee Yoo-jae and Kim Woo-cheol [79], it was evaluated with four items: 'the facilities of my main bank's branches are clean', 'my main bank's branches are a space where I want to stay for a long time', 'my main bank's branches are generally convenient to use', and 'my main bank's branches are generally aesthetically pleasing'.

3.3.6. Digital Quality

Digital quality was subdivided into 'system quality', 'digital information quality', and 'service quality', and the measurement items were based on the studies of Lee, Kwak, and Lee; Hwang et al.; and Kim [61,85,110,111]. First, system quality was categorized into accessibility, security, and ease of use, while digital information quality was categorized into sufficiency, diversity, up-to-dateness, accuracy, and usefulness. Service quality was categorized into customer support, customization, and reliability. Conceptual definitions of variables and measurements were as follows (see Table 2).

Table 2. Conceptual definition and measure of digital quality.

Factor		Conceptual Definition	Measure
System Quality	Accessibility	Digital service access speed; always use, degree of information acquisition	The main bank's digital services (internet, mobile, platform) have fast access. The main bank's digital services (Internet, mobile, platform) are always available. The main bank's digital services (Internet, mobile, platform) are easy to obtain information.
	Security	System safety, privacy, degree of security	The main bank's digital services (Internet, mobile, platform) are safe from hacking. The main bank's payment and payment methods are highly secure. The main bank's security system is superior to that of other banks.
	Ease of Use	Simplicity, degree of affinity with using system	The main bank's digital services (Internet, mobile, platform) are easy to use. The main bank's digital services (Internet, mobile, platform) are simple to use. The main bank's digital services (Internet, mobile, platform) provide a user-friendly interface (system, design).
Information Quality	Sufficiency	Sufficiency of necessary information	The main bank's digital services (internet, mobile, platform) provide sufficient information about products (e.g., savings accounts, loans).
	Diversity	Diversity of necessary information	The main bank's digital services (Internet, mobile, platform) provide a variety of information about products (e.g., savings, loans).
	Freshness	Freshness of necessary information	The main bank's digital services (internet, mobile, platform) provide up-to-date information on products (e.g., savings accounts, loans).
	Accuracy	Accuracy of necessary information	The main bank's digital services (Internet, mobile, platform) provide accurate information about products (deposits, savings, loans).
	Usefulness	Usefulness of necessary information	The main bank's digital services (Internet, mobile, platform) provide useful information about products (deposits, savings, loans).
Service Quality	Customer Support	Prompt response, correct response, and sincere answers (communication) to customer requests	The main bank's digital solutions (chatbots, robo-advisors) respond quickly to my requests. The main bank's digital solutions (chatbot, robo-advisor) respond to my request with accurate content. The main bank's digital solutions (chatbots, robo-advisors) faithfully answer my requests.
	Customization	Customized products for customer preferences and interest, service provided such as information, etc.	The main bank's digital solutions (chatbot, robo-advisor) provide customized services tailored to my interests. The main bank's digital solutions (chatbot, robo-advisor) provide differentiated services according to my conditions and circumstances. The main bank's digital solutions (chatbot, robo-advisor) provide customized services according to my knowledge (skill) level.
	Reliability	Delivering promises, solutions, and defined services to customer needs	The main bank's digital solutions (chatbot, robo-advisor) solve my problem. The main bank's digital solutions (chatbot, robo-advisor) execute within the promised time. The main bank's digital solutions (chatbot, robo-advisor) provide the promised service.

Source: Created by the author.

3.3.7. Customer Satisfaction

Customer satisfaction is measured by 'satisfaction' [18] or 'pleasure' [101,121]. In this study, the concept of customer satisfaction is defined as a combination of two concepts, 'satisfaction' and 'consumption emotion'. Expectations are satisfied when the performance actually obtained compares with or exceeds the result expected in advance, and the degree to which the performance of service attributes satisfies the needs of the service consumer determines the degree of customer satisfaction. On the other hand, among the emotional factors induced by consumers in the consumption situation, Richins [122] selected two emotional factors close to customer satisfaction to measure consumption sentiment.

Customer satisfaction is defined as 'a customer's level of satisfaction with the relationship between a brand and a customer' and is expressed as the sum of perceived quality and perceived loss from the service experience. In terms of operational definition, customer satisfaction is transaction-specific customer satisfaction, which is the result of evaluation by individual transactions based on the expectation–incongruity paradigm, and cumulative (overall) customer satisfaction is accumulated through individual satisfaction experiences for individual transactions [123].

The measurement items for customer satisfaction adopted in this study are based on the research of Oliver [124], Ragunathan and Irwin [98], and Lee and Ra [125]. Customer satisfaction was evaluated with four items: 'Using the main bank usually meets my expectations', 'my main bank's service is more satisfactory than other bank services', 'my overall level of satisfaction with my main bank's service is high', and 'transactions with my main bank are fun and enjoyable' to reflect both customers' expectations and needs.

3.3.8. Brand Loyalty

This paper focuses on behavioral and attitude loyalty, which are among the three customer loyalty concepts of Bowen and Chen [100]. Attitude loyalty encompasses the customer's preference for the company [126] and the customer's continuous good relationship with the company. Consistency of preference to maintain a relationship was measured [127]. In addition, behavioral loyalty was measured by repurchase intention to continue doing business with the company [128] and positive word of mouth by recommending the bank to people around them [13].

Brand loyalty is defined as a 'customer's favorable attitude, word-of-mouth, and repeated purchase behavior', based on the research of Sirgy and Samli [129] and Cho and Lim [117]. It was evaluated with four items: 'I have a very favorable (good) attitude towards my main bank', 'I once told others that my main bank is a good bank', 'I have many other banks to choose from, but I tend to use my main bank', and 'I will continue to use my main bank'.

3.3.9. Uncertainty

Uncertainty was composed of two factors: 'technical uncertainty' and 'market uncertainty'. Technological uncertainty is 'the degree of change and unpredictability of technology', and market uncertainty is 'the degree of change and unpredictability of the market (customer)'. Research items derived from previous studies, such as that of Lisboa, Skarmeas, and Lages [130], were modified and supplemented. Finally, uncertainty was evaluated with four items: 'it is very difficult to predict technological changes applied to the banking industry', 'technologies in the banking industry are changing rapidly', 'it is very difficult to predict the changing needs of customers in the banking industry', and 'the needs of customers in the banking industry are changing rapidly'.

3.4. Research Tool

The empirical analysis of this study applied structural equation analysis using the Smart-PLS analysis tool. Structural equation analysis is a statistical technique for measuring abstract concepts presented by researchers and testing causal relationships, including covariance and correlation between latent variables [131]. PLS (Partial Least Squares), a technique of structural equation modelling, is not limited by the number of variables, can be analyzed regardless of sample size, and is a useful method even if the accuracy of surveys and measurements is somewhat low [132]. It can also be used when the sample is not sufficient or the theoretical basis is not sufficient [133]. The sample to be analyzed in this study was 406 copies, and the Smart-PLS analysis method was used in consideration of the view that the sample may be somewhat insufficient and that there are insufficient previous studies on digital quality measurement items in the banking industry.

3.5. Sample Design and Data Collection

In this study, a survey was conducted from February 2023 to March 2023 using the Google survey system targeting 406 Korean customers who use banks. Due to difficulties in conducting a local survey, the survey was conducted focusing on cities rather than provinces. Since the proportion of a specific age group in the initial survey sample was very high, we used the quota sampling method to intentionally keep the survey subjects evenly distributed; thus, some of that sample was removed to balance with other age groups. A total of 406 questionnaire responses were collected. In addition, in order to increase the reliability of the survey, face-to-face interviews were conducted with some of the respondents, and the contents of the interviews were used to reconfirm the results of the statistical empirical analysis. In addition, the reliability of the survey was increased by removing samples of insincere survey responses, and finally, 330 copies were used as analysis data. The main demographic characteristics of this study sample are shown in Table 3.

Table 3. Demographic characteristics.

Gender	Male	256 people (63.2%)	Job	Employee	79 people (19.4%)
	Female	150 people (36.8%)		Public officer	13 people (3.2%)
Age	20–29	174 people (42.8%)		Under-graduate Post-graduate	166 people (40.8%)
	30–39	30 people (7.4%)		Business men	49 people (12%)
	40–49	35 people (8.6%)		Professional	50 people (12.3%)
	50–59	89 people (21.9%)		Service sector	11 people (4.4%)
	More than 60	76 people (18.7%)		Housewife	7 people (1.7%)
Education	High school students	3 people (0.7%)		Agriculture, fishery, forestry, animal husbandry	4 people (1%)
	Graduate from high school	42 people (10.3%)		Unemployed	21 people (5.2%)
	Graduate from college (enrolled)	25 people (6.1%)		Other	18 people (4.4%)
	Graduate from university (enrolled)	242 people (59.5%)	Average monthly income	<GBP 1200	183 people (45%)
	Post-graduate (enrolled)	95 people (23.3%)		GBP 1200~1800	49 people (12%)
				GBP 1800~2400	45 people (11.1%)
				GBP 2400~3000	38 people (9.3%)
				>GBP 3000	92 people (22.6%)

Source: Created by the author.

By gender, there were 256 males (63.2%) and 150 females (36.8%). In terms of age, the distribution was as follows: 174 individuals aged 20–29 (42.8%), 30 aged 30–39 (7.4%), 35 aged 40–49 (8.6%), 89 aged 50–59 (21.9%), and 76 aged 60 or older (18.7%). Regarding education level, there were 3 middle or high school students (0.7%), 42 individuals with a high school diploma or less (10.3%), 25 junior college students or graduates (6.1%), 242 college students or graduates (59.5%), and 95 graduate students or graduates (23.3%). By occupation, there were 79 office workers (19.4%); 13 civil servants (3.2%); 166 college or graduate students (40.8%); 49 self-employed or business owners (12%); 50 professionals (12.3%); 11 service workers (2.7%); 7 housewives (1.7%); 4 individuals in agriculture, fishery, forestry, or livestock (1%); 21 unemployed (5.2%); and 18 in other occupations (4.4%). In terms of average monthly income, 183 people (45%) earned less than GBP 1200, 49 (12%) earned between GBP 1200 and 1800, 45 (11.1%) earned between GBP 1800 and 2400, 38 (9.3%) earned between GBP 2400 and 3000, and 92 (22.6%) earned GBP 3000 or more.

4. Results

4.1. Model Evaluation and Verification

Prior to the analysis, the research model was evaluated. Reliability and conceptual validity (convergent validity) tests and multicollinearity analysis between and among variables were conducted, and discriminant validity was additionally tested.

4.1.1. Multicollinearity: Pearson Correlation and Variance Inflation Factor (VIF)

Each of the items constituting the concept must have independence, and if the degree of correlation between these items is high and multicollinearity exists, the variance value of the regression coefficient expands and the standard error increases, causing reliability problems in the estimated regression coefficient. This can distort the results of an empirical analysis [133]. To verify the results, this study conducted a multicollinearity test between factor items through Pearson's correlation coefficient and VIF values.

In the Pearson's correlation coefficient test, the cut-off value is 0.7 (or more), which indicates that there is no multicollinearity problem (see Table 4). On the other hand, if the VIF value is 5 or more, there is a potential collinearity problem [134]; beyond this, a conservative view is that multicollinearity needs to be suspected when the mean VIF value of each item exceeds 3.3 [135,136]. In this study, as a result of verification, the maximum VIF value was 5.539, the minimum value was 1.147, and the average value was 2.874, so it is determined that there was no problem of multicollinearity for any items.

Table 4. Pearson's correlation.

Variables		(1)	(2)	(3)	(4)	(5)	(7)	(8)	(9)	(10)	(11)	VIF
Brand Awareness (1)		1.000										1.363
Brand Availability (2)		0.541	1.000									1.503
Brand Image (3)		0.507	0.610	1.000								2.210
Customer Orientation (4)		0.409	0.540	0.653	1.000							3.939
Physical Quality (5)		0.412	0.562	0.604	0.645	1.000						1.887
Digital Quality	System Quality (6)	0.416	0.527	0.539	0.542	0.591	1.000					3.356
	Information Quality (7)	0.387	0.509	0.569	0.614	0.613	0.657	1.000				4.100
	Service Quality (8)	0.368	0.464	0.530	0.591	0.621	0.598	0.677	1.000			4.227
Customer Satisfaction (9)		0.427	0.541	0.566	0.597	0.531	0.574	0.569	0.590	1.000		3.217
Brand Loyalty (10)		0.464	0.531	0.531	0.642	0.646	0.586	0.595	0.582	0.622	1.000	2.065

Source: Created by the author.

4.1.2. Reliability and Convergent Validity: Cronbach's Alpha, Synthetic Reliability, Average Variance Extracted (AVE)

The suitability evaluation of the research model started with a factor analysis. This exploratory factor analysis was used to check whether individual measurement items were properly grouped into each construct (reliability and conceptual validity or convergent validity) in the developed measurement tool (measurement model). The analysis also consisted of a confirmatory factor analysis to confirm adequacy, that is, discriminant validity. In this study, PLS (Partial Least Squares) [137], which can simultaneously check factor analysis to measure variables and path modeling to explain the relationship between variables, was used as an analysis tool. PLS has no constraints on variables, residuals, or normal distribution, even when the model cannot be generalized due to a problem of poor fit due to a complex model and an increase in error values due to many variables [138].

A factor analysis was conducted on the 11 constructs and 58 measurement items used in this study (see Appendix B, Table A5). A factor is a collection of variables with high correlation coefficients, and it divided into a small number of variable groups. What is important in factor analysis is that the contents of items are well grouped into constructs, which are theoretical subfields. The judgment on this is made by the factor loading, which indicates the degree of correlation between each variable and the factor. The cut-off value of the factor loading should be 0.4 or more at a conservative level, and if it is 0.5 or more, it is considered to have very high significance. As a result of the analysis, it was confirmed that the factor loading values of the measurement items constituting each concept in this study were over 0.5.

As shown in Table 5, the measurement result of synthetic reliability was found to satisfy the criterion of 0.7 or higher, as suggested by Nunnally and Bernstein [139]. In addition, the average variance extraction value presented as a criterion for judging convergent validity satisfied all criteria of 0.5 or higher, as suggested by Fornell and Larcker [140] and Chin [141]. As a result, the convergent validity that connects the variables and factors input into this study can be judged to be valid. At the same time, the variables used to measure all constructs satisfied the Cronbach's alpha value of 0.7 or [139,142], ensuring high reliability. On the other hand, when Cronbach's alpha exceeds 0.95, there is concern about the possibility of 'common method bias' [143]. The Cronbach's alphas of the latent variables input into this research model were all less than 0.95, confirming that there is no concern about same-method bias.

Table 5. Reliability and validity.

Items		Cronbach's Alpha	rho_A	Synthetic Reliability	Average Variance Extracted (AVE)
Brand		0.711	0.714	0.813	0.569
Availability		0.736	0.770	0.829	0.551
Customer		0.932	0.934	0.949	0.787
p.Qaulity		0.860	0.867	0.900	0.643
Physical		0.830	0.843	0.887	0.662
Digital Quality	System	0.934	0.937	0.945	0.657
	Information	0.945	0.945	0.958	0.820
	Service	0.959	0.959	0.965	0.753
Satisfaction		0.925	0.926	0.947	0.817
Loyal		0.830	0.854	0.886	0.662

Source: Created by the author.

In addition, the average variance extracted (AVE) for the constructs was found to be more than 0.5, the standard value [140,141], and the Cronbach's alpha value was also confirmed to be more than 0.7, the standard value. On the other hand, the composite reliability was also shown to be 0.7 or higher [139], which is the standard value. Therefore, it can finally be judged that the convergent validity of this research model has been proven.

4.1.3. Discriminant Validity: Fornell–Larcker [130] Criteria

Since independence must be secured between constructs, discriminant validity was performed to confirm this. The discriminant validity test method is generally tested with two indices. First of all, the outer loading value associated with the construct to which each of the measurement items constitutes one construct is the cross-loading associated with all constructs [133,143]. As an alternative to complement this, the conservative method proposed by Fornell and Larcker [140] was used. This method checks whether the square root of the average variance extracted (AVE) value of each construct is greater than the correlation value between variables [140]. This study conducted discriminant validity using the above two methods. First, as a result of comparing single-loading and cross-loading, the single-loading of all items was larger than the cross-loading.

Second, we looked at the condition that the square root of AVE of all variables should be greater than the correlation coefficient [140]. As can be seen in Table 6, the square root value (shaded) of the variance extraction value (AVE) presented on the diagonal line is greater than the correlation coefficient with other variables presented below the diagonal line. This means that the variance shared by the relevant measurement items of each construct was greater than the variance shared by the other constructs [140,144,145]. Finally, all the constructs introduced in this study can be judged to have sufficient discriminant validity conditions.

Table 6. Discriminant validity verification (Fornell–Larcker method).

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Availability (1)	0.743									
Brand (2)	0.541	0.685								
Customer (3)	0.540	0.409	0.887							
Information (4)	0.509	0.387	0.614	0.905						
Loyal (5)	0.531	0.464	0.642	0.695	0.814					
p.Quality (6)	0.610	0.507	0.653	0.669	0.731	0.802				
Physical (7)	0.562	0.412	0.745	0.613	0.646	0.704	0.814			
Satisfaction (8)	0.541	0.427	0.697	0.769	0.822	0.766	0.731	0.904		
Service (9)	0.464	0.368	0.591	0.777	0.682	0.630	0.621	0.790	0.868	
System (10)	0.527	0.416	0.642	0.757	0.686	0.739	0.691	0.774	0.698	0.811

Source: Created by the author.

4.2. Robustness Test

To ensure the consistency of the research results, this study conducted a series of robustness tests, such as the non-response bias test, common method bias test, bootstrapping, Sobel test, and f2 test.

4.2.1. Non-Response Bias: Armstrong and Overton [146] *t*-Test

As suggested by Armstrong and Overton [146], in order to identify potential non-response bias, the data collected in the first half of the survey period and the data collected in the second half were used for 25% of the main variables. A *t*-test was conducted to confirm the difference in measurement items. As a result of the analysis, there was no significant difference in the average value of the measured variables by group between the first half and the second half of responders in the 95% confidence interval. Therefore, it can be judged that there was no non-response bias in the questionnaire data used in this study.

4.2.2. Common Method Bias: Check for Over/Under Estimation Errors

In the case of survey data, in the process of expressing the "perceived" values of the respondents rather than the empirical values for the survey items, we used a systematic error in which the correlation between variables was overestimated, i.e., the common method bias may be latent. Overestimation can cause an error that makes a statistically nonsignificant relationship significant [147,148] or a Type II error by reducing the correlation [146]. Ways to check the common method bias are to see whether all items are grouped into one factor (construct) or to see whether a particular dominant factor (construct) explains most of the total variance [149–152] (see Tables 7 and 8). Specifically, Harman's Single-Factor Test [153] is generally and frequently used. In this study, this method was also adopted to test the problem of the common method bias.

Table 7. Total variance.

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative (%)	Total	% of Variance	Cumulative (%)
1	7.220	42.470	42.470	7.220	42.470	42.470
2	1.625	9.557	52.027	1.625	9.557	52.027
3	1.178	6.927	58.954	1.178	6.927	58.954
4	0.998	5.869	64.822	0.998	5.869	64.822s

Source: Created by the author.

Table 8. Bootstrapping (a) of basic model.

Hypotheses	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	*p* Values
Brand Awareness → Satisfaction	−0.001	0.000	0.030	0.019	0.985
Availability → Satisfaction	−0.012	−0.010	0.034	0.341	0.733
Customer → Satisfaction	0.104	0.105	0.047	2.207	0.027
Information Quality → Satisfaction	0.136	0.137	0.049	2.780	0.005
Perceived Quality → Satisfaction	0.226	0.224	0.053	4.286	0.000
Physical → Satisfaction	0.132	0.132	0.040	3.340	0.001
Satisfaction → Loyalty	0.822	0.823	0.019	43.022	0.000
Service Quality → Satisfaction	0.308	0.306	0.043	7.191	0.000
System Quality → Satisfaction	0.137	0.138	0.047	2.883	0.004

Source: Created by the author.

4.2.3. Bootstrapping

Since PLS-SEM does not assume that the data form a normal distribution, bootstrapping is required to proceed with the significance test of coefficients. In addition, bootstrapping was performed to check the consistency of the analysis results of this study (see Tables 8–10). Bootstrapping does not require assumptions about the distribution of variables or the standard distribution of statistics, and it can be applied to a small sample size like this study using 526 survey data, so it is methodological in the case of statistical processing with a PLS-SEM model. Therefore, the validity is appropriate [154]. Therefore, as an analysis option for the PLS-SEM model in this study, the standard error of the path coefficient was estimated by specifying the number of sampling iterations as 5000 for more rigorous processing and by repeating non-parametric bootstrapping 5000 times. Thereafter, the significance of the pathway was confirmed by a *t*-test.

Table 9. Bootstrapping (b) of moderator model (uncertainty as a moderator).

Hypotheses	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	*p* Values
Brand Awareness → Satisfaction	−0.003	−0.003	0.036	0.090	0.928
Availability → Satisfaction	−0.008	−0.007	0.039	0.198	0.843
Customer → Satisfaction	0.093	0.094	0.049	1.882	0.060
Information Quality → Satisfaction	0.125	0.126	0.049	2.570	0.010
Perceived Quality → Satisfaction	0.252	0.248	0.061	4.133	0.000
Physical → Satisfaction	0.142	0.141	0.042	3.391	0.001
Satisfaction → Loyalty	0.749	0.750	0.027	27.909	0.000
Service Quality → Satisfaction	0.303	0.300	0.044	6.899	0.000
System Quality → Satisfaction	0.152	0.154	0.052	2.953	0.003
Uncertainty → Satisfaction	0.218	0.222	0.072	3.046	0.002
Uncertainty × System Quality → Satisfaction	−0.150	−0.138	0.109	1.373	0.170
Uncertainty × Information Quality → Satisfaction	0.090	0.088	0.102	0.881	0.378
Uncertainty × Service Quality → Satisfaction	0.027	0.015	0.096	0.280	0.780

Source: Created by the author.

Table 10. Bootstrapping (b) of moderator model (digital quality as a moderator).

Hypotheses	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	*p* Values
Brand Awareness → Satisfaction	−0.003	−0.007	0.036	0.090	0.928
Availability → Satisfaction	−0.008	−0.007	0.039	0.198	0.843
Customer → Satisfaction	0.093	0.094	0.049	1.882	0.060
Information Quality → Satisfaction	0.125	0.126	0.049	2.570	0.010
Perceived Quality →Satisfaction	0.252	0.248	0.061	4.133	0.000
Physical → Satisfaction	0.142	0.141	0.042	3.391	0.001
Satisfaction → Loyalty	0.749	0.750	0.027	27.909	0.000
Service Quality → Satisfaction	0.303	0.300	0.044	6.899	0.000
System Quality → Satisfaction	0.152	0.154	0.052	2.953	0.003
Uncertainty → Satisfaction	0.218	0.222	0.072	3.046	0.002
System Quality × Uncertainty → Satisfaction	−0.150	−0.138	0.109	1.373	0.170
Information Quality × Uncertainty → Satisfaction	0.090	0.088	0.102	0.881	0.378
Service Quality × Uncertainty → Satisfaction	0.027	0.015	0.096	0.280	0.780

Source: Created by the author.

4.2.4. Verification of Mediation Effect (Sobel Test)

In order to test the mediation effect of customer satisfaction, Sobel [155] verification was conducted using the multivariate delta method formula for the double mediation effect.

$$z = ab/\sqrt{a^2S_b^2 + b^2S_a^2}$$

In this case, a and b are the coefficients of paths a and b, and Sa and Sb are the standard errors of the corresponding paths, respectively. As shown in Table 11, customer satisfaction mediates the relationship between brand image, customer orientation, physical environment, digital quality (system quality, information quality, service quality), and brand loyalty, which showed statistical significance among service quality. The effect was found to be statistically significant in all pathways.

Table 11. Mediator test (Sobel test).

Path	Sobel Test (Z)	p (Two-Tailed)	S.D. (Satisfaction)
Image → Satisfaction → Loyalty	4.24 †	0.000	0.053
Customer Orientation → Satisfaction → Loyalty	2.20 **	0.027	0.047
Physical → Satisfaction → Loyalty	3.29 ***	0.001	0.040
System → Satisfaction → Loyalty	2.91 ***	0.003	0.047
Information → Satisfaction → Loyalty	2.76 ***	0.005	0.049
Service → Satisfaction → Loyalty	7.06 †	0.000	0.043

Notes: S.D. stands of Standard Deviation; S.D. of the relationship between satisfaction and loyalty is 0.019; † $p < 0.001$, *** $p < 0.01$, ** $p < 0.05$. Source: Created by the author.

4.2.5. Effect Size: f^2 Test

The f^2 test evaluates whether the removed constructs (knowledge sharing and knowledge creation, which are parameters in this article) actually have an impact on endogenous constructs. To this end, we used a method of comparing the R^2 value when a specific exogenous construct is removed from the model and the R^2 value when it is included. The standard values for determining the f^2 effect size were 0.02, 0.15, and 0.35 [134], representing small, medium, and large effects, respectively (see Table 12).

Table 12. Effect size (f2 test result).

Variables	F2 (Satisfaction)	F2 (Loyalty)
Brand Awareness	0.000	-
Brand Availability	0.000	-
Brand Image	0.079	-
Customer Orientation	0.021	-
Physical Quality	0.028	-
System Quality	0.027	-
Information Quality	0.026	-
Service Quality	0.158	-
Customer Satisfaction	-	2.083

Source: Created by the author.

The f^2 values derived from the calculation formula indicate predictive accuracy, as all antecedent variables—except for brand awareness and brand availability, which did not

show a statistically significant relationship with customer satisfaction—exhibited values greater than 0. Furthermore, the effect size analysis revealed that all values were greater than 0.02, indicating medium effect sizes. Consequently, the structural model including customer satisfaction as a mediating variable can be evaluated as superior to the model without it.

4.3. Results

4.3.1. Empirical Results

As a result of the analysis, it was found that brand awareness and brand availability (including economic quality) had no statistically significant relationship with customer satisfaction in the study of brand equity components in the banking industry (see Figure 2). In contrast, brand image, customer orientation, physical quality, and digital quality were found to have a statistically significant positive (+) relationship with customer satisfaction. In particular, in the case of digital quality, all three sub-constituent concepts of digital quality—system quality, information quality, and service quality—were found to have a statistically significant positive (+) relationship with customer satisfaction. In addition, customer satisfaction was found to have a positive (+) effect on brand loyalty.

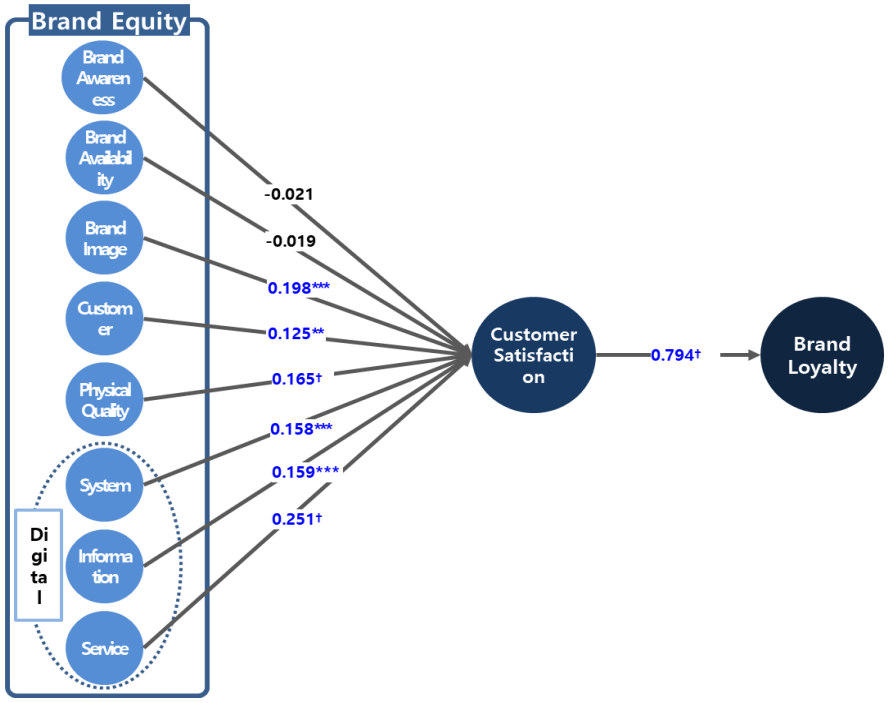


Figure 2. Results (a) of basic model. Notes: $p < 0.001$ †, $p < 0.01$ ***, $p < 0.05$ **. Source: Created by the author.

Regarding the moderating effect, uncertainty did not have a statistically significant moderating effect on the relationship between digital quality and customer satisfaction (see Figure 3). It was found that the moderating effect of uncertainty had no significant effect on all three components of digital quality. On the other hand, digital quality also did not have a significant moderating effect on the relationship between uncertainty and customer satisfaction (see Figure 4). None of the components of digital quality showed a moderating effect on the relationship between uncertainty and customer satisfaction.

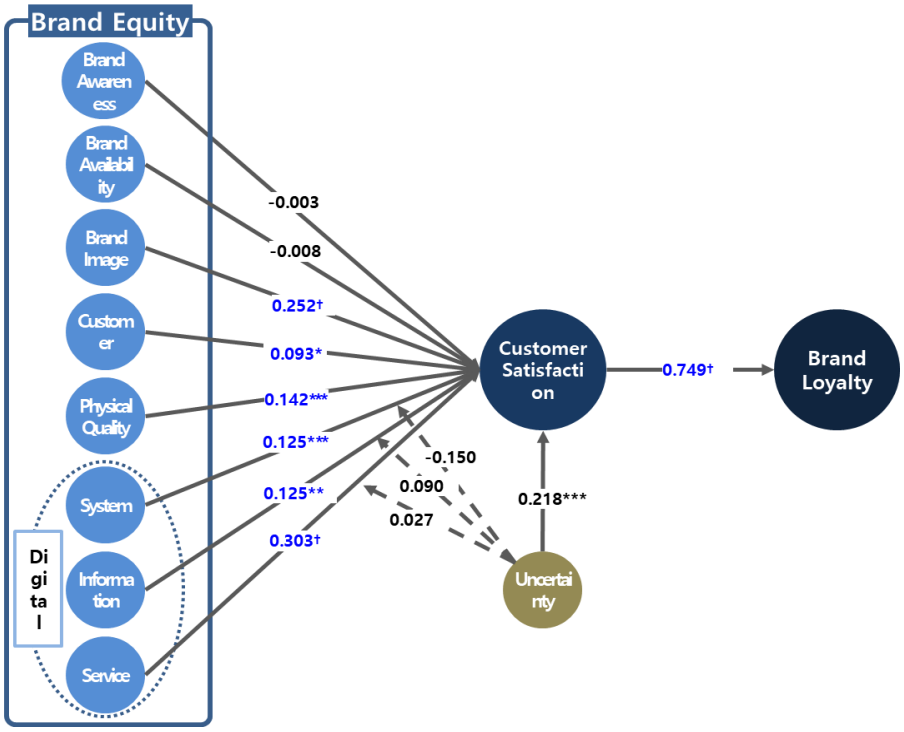


Figure 3. Results (b) of moderator model (uncertainty as a moderator). Notes: $p < 0.001$ †, $p < 0.01$ ***, $p < 0.05$ **, $p < 0.1$ *. Source: Created by the author.

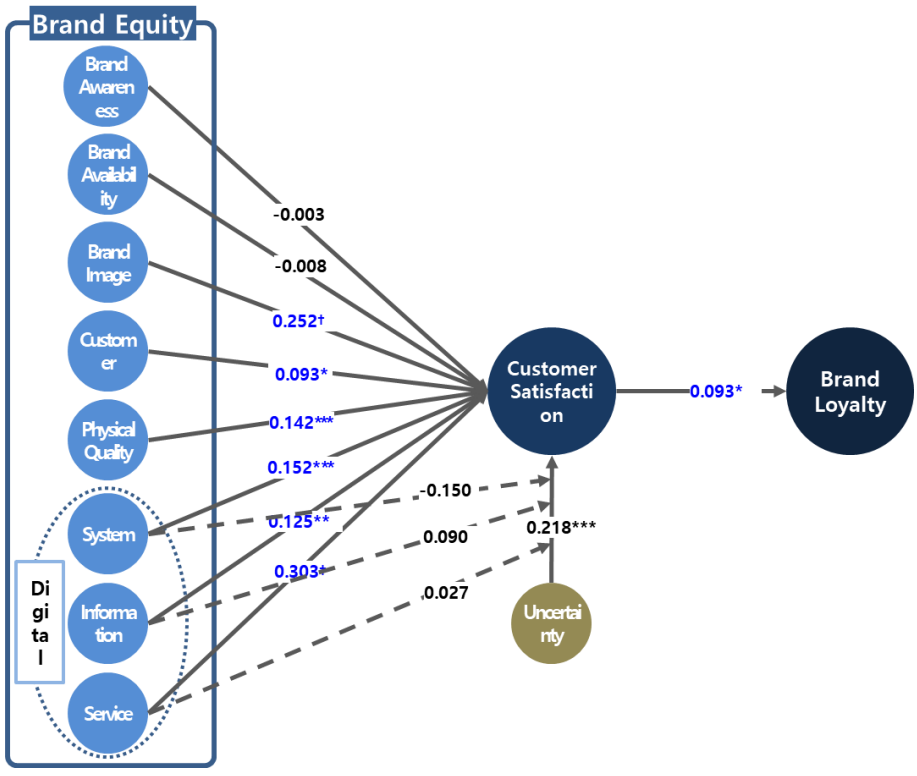


Figure 4. Results (c) of moderator model (digital quality as a moderator). Notes: $p < 0.001$ †, $p < 0.01$ ***, $p < 0.05$ **, $p < 0.1$ *. Source: Created by the author.

4.3.2. Summary of Hypothesis Testing

The results of hypothesis verification based on the results of the empirical analysis of this study are summarized in Table 13. Hypotheses 1-1, 1-2, and 7-2 were not supported, but all other hypotheses were supported.

Table 13. Brief summary of results.

Hypothesis	Path	Result
H1-1 (+)	Brand Awareness → Customer Satisfaction → Customer Loyalty	Not Supported
H1-2 (+)	Brand Image → Customer Satisfaction → Customer Loyalty	Not Supported
H2 (+)	Brand Availability → Customer Satisfaction → Customer Loyalty	Supported
H3 (+)	Customer Orientation → Customer Satisfaction → Customer Loyalty	Supported
H4 (+)	Physical Quality → Customer Satisfaction → Customer Loyalty	Supported
H5-1 (+)	System Quality → Customer Satisfaction → Customer Loyalty	Supported
H5-2 (+)	Information Quality → Customer Satisfaction → Customer Loyalty	Supported
H5-3 (+)	Service Quality → Customer Satisfaction → Customer Loyalty	Supported
H6 (+)	Customer Satisfaction → Customer Loyalty	Supported
H7-1 (+)	Uncertainty → Customer Satisfaction	Supported
H7-2 (+)	Uncertainty × Digital Quality → Customer Satisfaction	Not Supported

Source: Created by the author.

4.4. Analysis of Partial and Full Mediation

Full mediation refers to the case where the direct path coefficient between the predictor variable and the dependent variable is not statistically significant, and only the indirect effect between the predictor variable and parameter-dependent variable is statistically significant [156]. Partial mediation exists when both the direct path coefficient between the predictor variable and the dependent variable and the indirect effect between the predictor variable, the parameter variable, and the dependent variable are statistically significant [157]. Looking at Table 14, in the partial model, the direct effect (0.087 **) between brand awareness and customer loyalty was found to be statistically significant, and in the full model, the direct effect (−0.021) between the two was not statistically significant. Therefore, it can be judged that there was no mediating effect.

Also, for the relationship between brand availability and customer loyalty, the direct effect (0.009) was found to be statistically nonsignificant in the partial model and −0.019 in the full model, similarly nonsignificant. Therefore, it can be judged that there was no mediating effect of customer satisfaction.

As for the relationship between brand image and customer loyalty, the direct effect (0.300 †) was found to be statistically significant in the partial model, and the direct effect (0.198 ***) was also found to be significant in the full model. Therefore, it was confirmed that customer satisfaction had a partial mediating effect.

As for the relationship between customer orientation and customer loyalty, the direct effect (0.122 **) was statistically significant in the partial model, and the direct effect (0.125 **) was found to be significant in the full model. Therefore, it was confirmed that customer satisfaction had a partial mediating effect.

As for the relationship between physical quality and customer loyalty, the direct effect (0.046) was not statistically significant in the partial model, whereas the direct effect (0.165 †) was statistically significant in the full model. Therefore, customer satisfaction was found to have a full mediating effect.

Regarding the relationship between system quality and customer loyalty, which are components of digital quality, the direct effect (0.059) was not statistically significant in the partial model, whereas the direct effect (0.158 ***) was statistically significant in the full model. Therefore, customer satisfaction was found to have a full mediating effect. As for the relationship between information quality and customer loyalty, the direct effect (0.156 ***) was found to be statistically significant in the partial model, and the direct effect (0.159 ***) was also found to be significant in the full model. Therefore, it was confirmed that customer satisfaction had a partial mediating effect. As for the relationship between service quality and customer loyalty, the direct effect (0.193 †) was statistically significant in the partial model, and the direct effect (0.251 †) was also found to be significant in the full model. Therefore, it was confirmed that customer satisfaction had a partial mediating effect.

Table 14. Analysis of partial mediation and full mediation.

Hypothesis	Path	Path Efficient	Result
H1-1	Brand Awareness → Customer Loyalty	0.087 **	Not Mediated
	Brand Awareness → Customer Satisfaction → Customer Loyalty	−0.021	
H1-2	Brand Image → Customer Loyalty	0.300 †	Partial Mediated
	Brand Image → Customer Satisfaction → Customer Loyalty	0.198 ***	
H2	Brand Availability → Customer Loyalty	0.009	Not Mediated
	Brand Availability → Customer Satisfaction → Customer Loyalty	−0.019	
H3	Customer Orientation → Customer Loyalty	0.122 **	Partial Mediated
	Customer Orientation → Customer Satisfaction → Customer Loyalty	0.125 **	
H4	Physical Quality → Customer Loyalty	0.046	Full Mediated
	Physical Quality → Customer Satisfaction → Customer Loyalty	0.165 †	
H5-1	System Quality → Customer Loyalty	0.059	Full Mediated
	System Quality → Customer Satisfaction → Customer Loyalty	0.158 ***	
H5-2	Information Quality → Customer Loyalty	0.156 ***	Partial Mediated
	Information Quality → Customer Satisfaction → Customer Loyalty	0.159 ***	
H5-3	Service Quality → Customer Loyalty	0.193 †	Partial Mediated
	Service Quality → Customer Satisfaction → Customer Loyalty	0.251 †	

Notes: $p < 0.001$ †, $p < 0.01$ ***, $p < 0.05$ **. Source: Created by the author.

5. Discussion

5.1. Theoretical Implications

This study identified the relationship between brand equity components in the banking industry through an empirical analysis using a survey. It is significant that the existing brand equity component model was applied to the banking industry to construct and verify the variables that can evaluate the perceived quality measurement items of the bank brand. This provides the following implications.

First, the basic logic that service quality plays an important role as an antecedent factor for customer satisfaction and loyalty in the structural relationship of service production and consumption was confirmed in the banking industry. In addition, from the fact that brand awareness and brand availability (including economic quality) were found to have no statistically significant relationship with customer satisfaction as an antecedent factor that determines customer satisfaction, structural changes in banking service quality, which has traditionally been regarded as important, are progressing. It can be confirmed that brand image, customer orientation, and the physical environment are the main determinants of

service quality rather than the bank's awareness or availability (including economic quality). This tells us that contemporary customers are oriented toward a new component of service quality, breaking away from the principle of survival and competition in the traditional banking industry, which is characterized by a great deal of failure. Although it is a new financial institution, rather than brand awareness and usability symbolized by business history and scale, the special appeal of brand image for consumers is becoming more important. Mobile digital specialized banks such as Kakao Bank, K Bank, and Toss Bank in Korea, which are surpassing the market competitiveness of general commercial banks, and new financial service institutions such as Monzo in Europe, ensure high security and safety while providing ease of use to meet the new needs of financial services customers.

On the other hand, in a rapidly changing business environment, companies must invest a lot of resources and strategies in innovation activities to secure competitiveness. A company's propensity to propose and implement new ideas, that is, innovation or technology orientation, promotes the launch of new and attractive products, services, and technologies in the global market [158]. Innovation plays a key factor in corporate competitiveness in securing competitive advantage, restructuring, and growing organizations [158]. Therefore, it can be said that a company's pursuit of innovation is the key to its continued survival and growth [158]. Accordingly, contemporary corporations and banks are actively trying to gain a competitive advantage in the new phase of innovation through digital transformation. However, keep in mind that innovation orientation or technology orientation based on digital technology or knowledge is an important driving force for securing competitive advantage, but if the customer orientation factor is overlooked, consumer needs cannot be met [159]. Failure to lead may result in competitors preempting consumers' unmet service needs, leaving the firm unable to develop new capabilities, or leaving the market [160]. As such, in order to improve competitive advantage performance, companies must recognize the importance of customer orientation and innovation orientation and strive to balance these characteristics at the organizational level.

Second, digital quality was found to have a statistically significant positive (+) effect on customer satisfaction for all three components of system, information, and service quality. On the other hand, none of the three components of digital quality showed a moderating effect on the relationship between customer satisfaction and uncertainty. In banking services, customers are interested in banks' system quality, such as accessibility, security, ease of use of the bank's digital services (internet, mobile, etc.), customer support, customerization, and reliability of digital solutions (chatbots, robo-advisors, etc.). It has been scientifically proven that quality is an important preceding variable for customer satisfaction. In addition, among the digital quality factors, information quality, such as sufficiency, variety, up-to-dateness, accuracy, and usefulness of information about products offered through banks' digital services, was also proven to be a preceding factor for customer satisfaction.

Nevertheless, it was confirmed that digital quality does not have a regulating function that strengthens the structural relationship or results in companies making more efforts to meet customer needs in a market with great uncertainty, which would ultimately increase customer satisfaction. It can be judged that the level of digital quality in banking services is still low and that the limitations of digital quality functions are exposed due to the nature of the banking industry compared to the manufacturing industry.

On the other hand, it is worth paying attention to the fact that traditional service quality components such as the physical environment still determine the quality of banking service perceived by customers, although the meaning of traditional banking service quality components such as brand awareness and brand availability is fading. While existing studies on bank service quality have focused on examining the positive function as an

antecedent factor of customer satisfaction and customer loyalty, research on contemporary bank service quality, where the landscape is changing with the development of technology representing the fourth industrial revolution, must adopt a dynamic approach that focuses on changes in the functions of individual components rather than generalization through consideration of the precedence relationship between service quality components and customer satisfaction. In the same context, the overall service quality experienced by customers through the physical environment and virtual channels is defined as multi-channel service quality, and includes face-to-face service channels such as bank branches and internet banking. Studies have also considered service channels through virtual routes such as app services [161].

The above research results go beyond the approach of separately studying the traditional face-to-face channel and the virtual channel, which has begun attracting attention, to stimulate theoretical research on the connectivity between the two channels [162]. Specifically, it is interesting to consider that customers' negative experiences in individual channels may be transferred to other channels. It also opens up opportunities for theoretical research on the effect of increasing customer value that multi-channel integration will bring [163].

Third, as a result of examining the two-sided function of uncertainty, uncertainty in the banking service market has a statistically significant positive (+) effect on customer satisfaction, but there is no moderating effect on the relationship between digital quality and customer satisfaction. The positive (+) relationship between uncertainty and customer satisfaction derived from the results of this study is based on the results of studies examining the positive functions of market competition or uncertainty [106,164,165]. Business environment dynamics (or environmental uncertainty) refers to the frequency, intensity, and degree of unpredictability in the environment, such as technology, competitors, and customers [166–169]. This manifests as uncertainty in the business environment, and this uncertainty requires organizations to have a more flexible structure [170,171]. More active efforts are made to acquire external resources and integrate them with internal resources to utilize them [49]. The competitive advantages and strategic choices that a company generates through this series of dynamic capabilities respond appropriately to the changing needs of customers, achieving a balance with technology orientation and enhancing customer orientation, thus strengthening customer satisfaction.

Contrary to this fact, some previous studies have considered the relationship between the negative function of uncertainty and its two-dimensional effect. There is also competing understanding [172,173] that DCs' efficiency and potential for competitive advantage are relatively reduced in highly dynamic environments. On the other hand, Schilke [107] revealed that uncertainty has an inverted U-shaped relationship with competitive advantage, which is an empirical indicator of corporate performance. From this fact, we can see that it is necessary to consider business environment dynamism from multiple dimensions. According to the one-dimensional understanding of the impact of environmental uncertainty, business environmental uncertainty is understood to reduce the resource efficiency and competitive advantage of a company [106,169,172,174] and is also understood to enhance the competitive advantage of resources [175].

Furthermore, it is also necessary to confirm whether different influences appear depending on the aspect or type of uncertainty. Corresponding to the fact that a company is aware of the possibility of changes that will occur in the future, Miller [30] and Werner, Brouthers and Brouthers [31], and Courtney et al. [176] suggested that the uncertainty we commonly refer to can be classified as 'perceived uncertainty'. On the other hand, Courtney et al. [176] defined 'residual uncertainty' by distinguishing factors that hinder the results from general uncertainty, even though all conceivable circumstances were controlled. In addition, Carson et al. [177] defined uncertainty as volatility in terms of external factors

and ambiguity in terms of internal factors. Uncertainty is also categorized as internal uncertainty. If basic data measured according to the concept of uncertainty in these different forms can be utilized, an in-depth examination of the two-faced function of uncertainty that appears in the service process will be possible. Additionally, this work is expected to open new horizons for research on banking service quality.

5.2. Managerial Implications

This study classifies customers' perceived quality into six factors and empirically analyzes the factors that affect customer satisfaction and brand loyalty. The following suggestions can be made to bank practitioners and executives regarding which factors should be the focus of management and improvement.

First, banks should comprehensively identify the process by which service quality is established to enable systematic and structured retail banking strategies that ensure customer satisfaction and loyalty, while putting digital quality into practice to meet customer needs and improve perceived quality. A detailed, comprehensive, and integrated strategy that can be applied is required. In order to have a sustainable competitive advantage in the retail financial service field, it is necessary to improve service quality as perceived by customers and strengthen digital service quality, which is currently being emphasized, to connect with practical customer satisfaction and trust, ultimately strengthening customer's continuous loyalty. In line with the era of digital transformation, banks dealing with customers' sensitive asset products should offer high-speed access to digital services, always-on use, ease of information acquisition, safe systems, protection of personal information, security, and a system that is easy and simple to use. Continuous investment and management affinity will be required.

Most banks, such as Kookmin Bank's Vivi, Liv Smart, Shinhan Bank's Aurora, Hana Bank's HAI, and Woori Bank's Webibot, are introducing two-way communication with customers through digital solutions (chatbots, robo-advisors). Accuracy, sincerity, providing customized and differentiated services, solving customer needs, and providing promised services are the prerequisites for customer satisfaction above all else. In particular, younger generations, who are emerging as new major customers, have a higher conversion rate to other brands when purchasing products and services than previous generations, making it difficult to rely on the reliability or brand loyalty inherent in the traditional banking industry. Therefore, it is necessary to improve services for digital solutions, provide customized and precise digital services, and accurately solve problems for each customer. Customer trust must be improved through immediate resolution of errors.

At the same time, banks need to strengthen multi-channel service quality management. Considering that digital quality and physical environmental factors are also important antecedents of customer satisfaction in banking services, banks should pursue multi-channel integration to strengthen the physical quality and digital channels at the same time. Customers of banking services will expect banks to provide comparable quality of service between traditional face-to-face and digital channels by managing the channels in an integrated way rather than focusing exclusively on one channel. If the service quality of offline channels and online channels is not balanced and shows bias, it is easy for consumers of bank services to switch to other banks. Considering that customer preference has a higher conversion potential than other industries, banks must understand that multi-channel service quality management is very meaningful in service marketing.

Second, it is necessary to strengthen technology orientation based on customer orientation in digital channels through the advancement of digital transformation. The main technologies of digital transformation in the banking industry are big data (target marketing, customized service provision, etc.), artificial intelligence (work automation, customer

credit evaluation, product recommendation, etc.), cloud (speed, product/service development, infrastructure environment establishment, etc.), and blockchains (authentication, payment/remittance, loan/investment, security, etc.). Digital transformation using these digital technologies is leading to changes in the value chain of existing industries as well as existing business processes. Process changes that enhance the efficiency and competitiveness of corporate operations, and optimization and reorganization of business models based on these changes, is possible [178].

In order to continuously increase profits, reduce costs, and increase customer satisfaction in the banking and other financial industries in the era of the fourth industrial revolution, the operating model must be fundamentally redesigned, and the core approach required for operating model innovation must be pursued in a holistic and continuous manner [83]. Therefore, it is important to define the success factor of digital transformation as SMACIT3 (Social, Mobile, Analytics, Cloud, Internet of Things) technology and apply related technology to attract customers to strengthen service and customer relationships [179].

Since customers do not visit banks frequently, a digital-based marketing strategy through channels such as the Internet, e-commerce, and smartphones is important for effective sales and marketing [180]. For the service industry, it is important to digitize and improve communication and contact points with customers [113].

Kakao Bank, K Bank, and Toss Bank in Korea are Internet-only banks that conduct transactions and communicate with customers online only without offline stores, and their market share is also on the rise. As such, even in the traditional banking industry, digital quality based on digital capabilities is important in transactions and interactions with customers. To this end, digital transformation of the banking industry is a prerequisite, and since 2016, when fintech corporate banks entered the financial industry in earnest, digital transformation has become a key strategic task for banks.

Third, it is necessary to change the old perception of uncertainty management. Companies react differently to changes in the same business environment. In general, while most banks view uncertainty as an object to eliminate or avoid, there are banks that see uncertainty as an opportunity. In general systems theory, the ambivalent response of objects to uncertainty is a very common phenomenon that can occur in society, because society has a continuum of feedback and the resulting organized complexity of subsystems [181–183].

To put the ongoing discussions together, there are conflicting interpretations of the function of uncertainty depending on the theories or perspectives applied by each study, so it is no longer practical to understand uncertainty only with fragmentary causal relationships. In particular, Courtney et al. [176] stated that controlling for one uncertainty does not mean that a firm has achieved certainty. Therefore, it is reasonable for a manager who aims to achieve maximum performance by securing corporate viability and management efficiency to take the positive side of uncertainty and understand the negative side as a factor to be managed.

As such, changes in banks' perception of uncertainty may lead to appropriate strategy selection. Strategies that banks can choose to respond to uncertainty include dynamic capacity and ambidextrous strategies. In order for global companies to turn the ambivalent effects of business environment uncertainty into their own opportunities, technological innovation and organizational innovation that can continuously secure a competitive advantage [184–187]. For this series of innovations, banks first need dynamic capabilities. Dynamic capability is defined as "the ability of a company to utilize or reorganize its internal and external resources and capabilities to respond to a rapidly changing environment" [188]. It is a series of knowledge management processes that acquire knowledge from outside and apply it to service innovation [189].

Dynamic capability is the restructuring of organizational capabilities appropriate to the environment in order to actively respond to environmental changes based on organizational system and strategy changes. Banks equipped with these dynamic capabilities utilize and explore internal and external resources in a way that is always responsive to technological changes and changing market needs [108], transform business models to meet opportunities, and reorganize and redeploy resources and capacities to be commensurate with the model. As a result, the ability to design and redesign an organizational architecture capable of seizing the opportunities of change and enabling short-term survival and long-term sustainable growth becomes excellent [190].

Dynamic capabilities [190] consist of three components: sensing, seizing, and reconfiguring. However, both customer orientation and technology orientation should also be considered as components, as a company must be capable of preparing sensitively for technological changes and changes in market needs. As shown in the results of this study, customer orientation, which consists of listening to the voices of customers, paying attention to their needs, providing accurate and relevant information to customers, and keeping promises with customers [191], is an important antecedent of customer satisfaction. By satisfying customers' needs better than competitors, banks can create competitive advantages that determine service quality and corporate performance for customer satisfaction [192].

On the other hand, according to March [108], who emphasized the active role of organizational learning from the perspective of organizational adaptation theory against the traditional environmental selection theory, companies seek a balance between exploitation and exploration activities in order to effectively manage uncertainty. In addition to customer orientation, banks require technology orientation to lead the future market, and technology orientation requires an ambidextrous strategy that simultaneously pursues short-term exploitation and long-term exploration activities. Unlike the general opinion [193–195] that corporate R&D investment activities have a positive (+) relationship with corporate performance, banks should have a different focus on R&D. The point is that the short-term and mid- to long-term performance implications of investment activities are complementary. Companies can improve their performance and realize sustainable growth when they balance short-term development activities with mid- to long-term research activities [100]. O'Reilly III and Tushman [196], Raisch et al. [186], Gibson and Birkinshaw [197], He and Wong [198], and Lubatkin et al. [199] understood this as an 'ambidexterity' strategy. Banks carry out organizational adaptation strategies through strategic choices that realize corporate performance improvement by utilizing existing routines and exploring new possibilities by pursuing exploitation and exploration activities in a balanced way [197,200,201]. Utilization innovation refers to innovation with low uncertainty, whose main purpose is to meet the needs of current customers and markets through cost improvement and process innovation in accordance with existing customer and market needs. In contrast, exploratory innovation refers to innovation with high uncertainty for the purpose of disruptive product innovation and creation of needs to create new markets and discover new customers [198,200]. The two-sided pursuit of innovation in utilization and exploration activities enables short-term development and mid- to long-term research activities, resulting in an ambidextrous organization that can realize both market survival and sustainable growth [196,197].

6. Conclusions

In this study, the perceived quality of the brand equity components of the banking industry was measured by six factors, and the digital quality factors were subdivided based on the age and importance of digital transformation. As a result of the study, it was found that brand awareness and brand availability among the perceived quality measurement

factors were not important antecedent determinants of customer satisfaction, while brand image, customer orientation, and physical quality were important antecedent determinants of customer satisfaction and had a positive effect on brand loyalty. Structural relationships could be identified. This study additionally considered how digital quality, which consists of system quality, information quality, and service quality, enhances brand loyalty through customer satisfaction. This study derived theoretical and practical implications. It is important for the banking industry to manage and strengthen digital quality capabilities due to the acceleration and necessity of today's digital transformation. This study subdivides digital quality into system quality, digital information quality, and service quality, and presents evaluation and measurement indicators.

Traditionally, brand awareness and brand availability have been recognized as key components of service quality that drive customer loyalty by enhancing customer satisfaction. In contrast, this study found that brand image, physical quality, and digital quality are brand equity factors that stimulate customer satisfaction. With the advent of the digital revolution, it appears that customer evaluations of service channels—particularly those based on software aspects such as digital service quality and brand image—are becoming more important than traditional brand equity components rooted in corporate size in the banking industry. Nevertheless, the continued significance of physical quality as a key determinant of customer satisfaction highlights that customers value both the physical environment and virtual (digital) channels. Therefore, the goal of banking services should not be to choose between the two but to effectively integrate both. This suggests the significance of multi-channel service quality management that simultaneously addresses both physical and digital service touchpoints.

Despite the research results and beneficial implications, the limitations of this study and future research directions are as follows. First, the method adopted in this study for the operational definition of service quality at the methodological level was conceptualized as consumers' perception of performance values for various attributes representing service quality. On the other hand, if we adopt the concept of comparing expectations and performance values for attributes, we wonder what changes will occur in the structural relationship between service quality and customer satisfaction and loyalty. Evaluating service quality is comparing consumers' perception of the service they actually received with the service they expected to receive [202]. In the same context, it is also interesting to compare and consider the difference between transaction-specific customer satisfaction measures and cumulative satisfaction measures based on the expectation–incongruence paradigm for customer satisfaction measures.

Second, research on the difference in quality perception according to individual characteristics needs to be added. In particular, additional research is needed on the difference in quality perception according to customer age. Regarding the improvement of quality perception according to the bank's digital conversion, it is expected that there will be a difference in quality perception according to the age of the customer. In this study, we targeted young people who are more likely to have access to financial services following digital transformation. Age has been treated as an important variable in customer characteristics in the case of service research. In the field of social psychology research, research results have indicated that the older the customer, the more efforts they make to collect information when making decisions [203]. Elderly people with relatively low cognitive abilities must use higher cognitive resources to gain access to innovative products or services and new distribution channels, and they perceive higher levels of risk or uncertainty [204].

Third, this customer survey was conducted on Korean bank users. Therefore, in-depth research will be possible if comparative studies are conducted on major countries such as the Asian region, the United States, the United Kingdom, China, and Japan. Additionally, the six perceived quality factors presented above do not reflect the whole of quality. Therefore, if the elements of perceived quality are subdivided and diversified in the future, it will be possible to develop this into a better study.

Fourth, with the development of Internet-only banks such as K-Bank, Kakao Bank, and Toss Bank, research is needed on customers using Internet-only banks, comparative studies with existing commercial banks, comparative studies between other industries or different industries, and on internal customers (employees). If these few things are supplemented, valid research can be done in this field, and it will be able to contribute more to the development of marketing.

Fifth, it is interesting to analyze the interaction between customer trust, satisfaction, and customer retention through the concept of 'switching cost' perceived by customers, which has recently attracted attention. The conversion cost incurred by moving to another bank instead of the existing main bank service is an important factor in the process of customer loyalty. Relationship quality is formed through the function of adjusting the relationship between perceived quality and repeated use of retail banking services.

Sixth, in terms of internal marketing, a study is needed on the effect of strengthening digital quality on the motivation, job commitment, and organizational commitment of non-customer members. A study should explore the structural relationship between digital quality and customer satisfaction and preference according to digital transformation in the banking industry. This could expand the horizon of research on relationship change.

Lastly, there has recently been interest in banks' strategic choices corresponding to changes in the perception of uncertainty, dynamic capabilities, ambidextrous strategies, and resilience of companies after the outbreak of the COVID-19 pandemic. A study on the strategies used to secure a competitive advantage also has value as a future research topic.

Only when these series of complementary studies are conducted can our understanding of the structural relationship between the antecedent factors of service quality in the banking industry and customer satisfaction and customer loyalty be complete. It is hoped that this study will serve as a primer to stimulate these future studies.

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Appendix A. Survey Questionnaire

Table A1. Brand equity.

Question	Not at All			Neutral			Very Much So
I can often come across advertisements and promotions of my main bank.	①	②	③	④	⑤	⑥	⑦
My main bank's brand is easy to remember.	①	②	③	④	⑤	⑥	⑦
My main bank's logo is easily distinguishable.	①	②	③	④	⑤	⑥	⑦
My main bank is a representative brand in the banking industry.	①	②	③	④	⑤	⑥	⑦
I am well informed of the characteristics of my main bank.	①	②	③	④	⑤	⑥	⑦
My main bank has more branches compared to competing brands.	①	②	③	④	⑤	⑥	⑦
My main bank offers more diverse financial products than other banks.	①	②	③	④	⑤	⑥	⑦
My main bank provides more advantageous (profitable) financial products than other banks.	①	②	③	④	⑤	⑥	⑦
My main bank's products are less risky than other banks.	①	②	③	④	⑤	⑥	⑦
My main bank has a sophisticated image.	①	②	③	④	⑤	⑥	⑦
My main bank gives me a sense of familiarity.	①	②	③	④	⑤	⑥	⑦
My main bank has a trustworthy image.	①	②	③	④	⑤	⑥	⑦
My main bank gives an image of leading the times.	①	②	③	④	⑤	⑥	⑦
My main bank is reputable.	①	②	③	④	⑤	⑥	⑦
The staff at my main bank's branches is friendly.	①	②	③	④	⑤	⑥	⑦
The staff at my main bank's branches is very responsive to questions.	①	②	③	④	⑤	⑥	⑦
The staff at my main bank's branches wants to know what customers want.	①	②	③	④	⑤	⑥	⑦
The staff at my main bank's branches offers good solutions to customers.	①	②	③	④	⑤	⑥	⑦
The staff at my main bank's branches takes care of my problems quickly.	①	②	③	④	⑤	⑥	⑦
The facilities of my main bank's branches are clean.	①	②	③	④	⑤	⑥	⑦
My main bank's branches are a space where I want to stay for a long time.	①	②	③	④	⑤	⑥	⑦
My main bank's branches are generally convenient to use.	①	②	③	④	⑤	⑥	⑦
My main bank's branches are generally aesthetically pleasing.	①	②	③	④	⑤	⑥	⑦

Table A2. Digital transformation (digital quality). Please respond to the survey by comparing your main bank's digital products and services with those of other banks based on the last three years.

Question	Not at All			Neutral			Very Much So
The main bank's digital services (internet, mobile, platform) have fast access.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are always available.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are easy to obtain information.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are safe from hacking.	①	②	③	④	⑤	⑥	⑦
The main bank's payment and payment methods are highly secure.	①	②	③	④	⑤	⑥	⑦
The main bank's security system is superior to that of other banks.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are easy to use.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are simple to use.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) provide a user-friendly interface (system, design).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are easy to use.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) are simple to use.	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) provide a user-friendly interface (system, design).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (internet, mobile, platform) provide sufficient information about products (e.g., savings accounts, loans).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) provide a variety of information about products (e.g., savings, loans).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (internet, mobile, platform) provide up-to-date information on products (e.g., savings accounts, loans).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) provide accurate information about products (deposits, savings, loans).	①	②	③	④	⑤	⑥	⑦
The main bank's digital services (Internet, mobile, platform) provide useful information about products (deposits, savings, loans).	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbots, robo-advisors) respond quickly to my requests.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) respond to my request with accurate content.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbots, robo-advisors) faithfully answer my requests.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) provide customized services tailored to my interests.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) provide differentiated services according to my conditions and circumstances.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) provide customized services according to my knowledge (skill) level.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) solve my problem.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) execute within the promised time.	①	②	③	④	⑤	⑥	⑦
The main bank's digital solutions (chatbot, robo-advisor) provide the promised service.	①	②	③	④	⑤	⑥	⑦

Table A3. Customer satisfaction and brand loyalty.

Question	Not at All			Neutral			Very Much So
Using the main bank usually meets my expectations.	①	②	③	④	⑤	⑥	⑦
My main bank's service is more satisfactory than other bank services.	①	②	③	④	⑤	⑥	⑦
My overall level of satisfaction with my main bank's service is high.	①	②	③	④	⑤	⑥	⑦
Transactions with my main bank are fun and enjoyable and reflect both my expectations and needs.	①	②	③	④	⑤	⑥	⑦
I have a very favorable (good) attitude towards my main bank.	①	②	③	④	⑤	⑥	⑦
I once told others that my main bank is a good bank.	①	②	③	④	⑤	⑥	⑦
I have many other banks to choose from, but I tend to use my main bank.	①	②	③	④	⑤	⑥	⑦
I will continue to use my main bank.	①	②	③	④	⑤	⑥	⑦

Table A4. Uncertainty.

Question	Not at All			Neutral			Very Much So
It is very difficult to predict technological changes applied to the banking industry.	①	②	③	④	⑤	⑥	⑦
Technologies in the banking industry are changing rapidly.	①	②	③	④	⑤	⑥	⑦
It is very difficult to predict the changing needs of customers in the banking industry.	①	②	③	④	⑤	⑥	⑦
The needs of customers in the banking industry are changing rapidly.	①	②	③	④	⑤	⑥	⑦

Appendix B

Table A5. Indicator loading and cross-loading.

Items	Brand	Availability	Customer	*p*.Quality	Physical	System	Information	Service	Satisfaction	Loyal
Brand1	0.723	0.365	0.273	0.346	0.289	0.296	0.277	0.278	0.289	0.319
Brand2	0.724	0.269	0.249	0.342	0.259	0.264	0.261	0.232	0.283	0.327
Brand3	0.745	0.323	0.273	0.296	0.270	0.246	0.234	0.274	0.307	0.325
Brand4	0.742	0.420	0.370	0.402	0.349	0.301	0.249	0.256	0.299	0.276
Brand5	0.769	0.470	0.226	0.346	0.235	0.315	0.305	0.211	0.279	0.338
Avail1	0.398	0.722	0.334	0.303	0.329	0.195	0.168	0.179	0.224	0.240
Avail2	0.445	0.847	0.447	0.505	0.495	0.460	0.428	0.401	0.464	0.420
Avail3	0.412	0.787	0.399	0.506	0.413	0.433	0.492	0.439	0.486	0.484
Avail4	0.372	0.794	0.423	0.450	0.418	0.406	0.327	0.279	0.355	0.373
Customer1	0.361	0.477	0.863	0.562	0.651	0.546	0.493	0.463	0.560	0.544
Customer2	0.390	0.489	0.912	0.576	0.678	0.579	0.541	0.510	0.609	0.596
Customer3	0.358	0.464	0.874	0.571	0.643	0.555	0.549	0.523	0.608	0.552
Customer4	0.354	0.501	0.896	0.615	0.666	0.580	0.576	0.554	0.658	0.585
Customer5	0.356	0.462	0.890	0.570	0.666	0.583	0.557	0.562	0.647	0.569
Image1	0.379	0.430	0.386	0.723	0.515	0.514	0.395	0.405	0.515	0.430
Image2	0.392	0.484	0.568	0.768	0.518	0.526	0.537	0.525	0.621	0.608
Image3	0.464	0.530	0.577	0.832	0.584	0.594	0.523	0.460	0.601	0.611
Image4	0.372	0.477	0.474	0.813	0.583	0.662	0.591	0.549	0.638	0.591
Image5	0.429	0.520	0.595	0.867	0.617	0.656	0.611	0.570	0.681	0.666

Table A5. *Cont.*

Items	Brand	Availability	Customer	*p*.Quality	Physical	System	Information	Service	Satisfaction	Loyal
Physic1	0.350	0.383	0.695	0.500	0.746	0.514	0.413	0.402	0.494	0.463
Physic2	0.323	0.459	0.560	0.569	0.844	0.568	0.492	0.492	0.559	0.480
Physic3	0.355	0.511	0.657	0.583	0.853	0.589	0.558	0.576	0.693	0.613
Physic4	0.316	0.462	0.527	0.632	0.808	0.574	0.513	0.528	0.607	0.526
System1	0.342	0.393	0.494	0.563	0.574	0.800	0.574	0.509	0.620	0.554
System2	0.339	0.337	0.495	0.477	0.489	0.723	0.498	0.472	0.564	0.464
System3	0.348	0.455	0.569	0.626	0.621	0.877	0.679	0.636	0.712	0.617
System4	0.315	0.462	0.539	0.631	0.545	0.793	0.597	0.544	0.558	0.491
System5	0.390	0.480	0.604	0.638	0.599	0.790	0.627	0.577	0.616	0.560
System6	0.344	0.485	0.546	0.652	0.587	0.749	0.589	0.560	0.594	0.547
System7	0.354	0.399	0.498	0.594	0.543	0.884	0.648	0.611	0.658	0.583
System8	0.309	0.412	0.463	0.578	0.515	0.846	0.636	0.581	0.657	0.579
System9	0.298	0.423	0.482	0.635	0.568	0.820	0.658	0.588	0.647	0.591
Infor1	0.329	0.446	0.526	0.614	0.550	0.703	0.888	0.660	0.678	0.608
Infor2	0.326	0.428	0.532	0.605	0.559	0.719	0.916	0.725	0.716	0.662
Infor3	0.384	0.476	0.529	0.600	0.525	0.681	0.920	0.714	0.698	0.630
Infor4	0.361	0.462	0.604	0.617	0.577	0.647	0.891	0.691	0.690	0.610
Infor5	0.351	0.496	0.587	0.592	0.561	0.676	0.912	0.728	0.696	0.634
Service1	0.339	0.412	0.582	0.596	0.569	0.690	0.727	0.859	0.704	0.639
Service2	0.322	0.426	0.546	0.568	0.550	0.663	0.700	0.878	0.696	0.587
Service3	0.326	0.441	0.557	0.584	0.566	0.641	0.694	0.846	0.670	0.605
Service4	0.290	0.396	0.466	0.520	0.478	0.596	0.675	0.900	0.683	0.579
Service5	0.298	0.385	0.452	0.504	0.543	0.585	0.645	0.871	0.679	0.584
Service6	0.262	0.370	0.426	0.480	0.473	0.565	0.657	0.869	0.658	0.551
Service7	0.321	0.401	0.489	0.541	0.579	0.624	0.659	0.882	0.698	0.550
Service8	0.352	0.369	0.527	0.552	0.521	0.522	0.628	0.839	0.660	0.587
Service9	0.358	0.422	0.564	0.574	0.567	0.563	0.685	0.867	0.716	0.645
Satis1	0.375	0.480	0.630	0.686	0.649	0.704	0.720	0.732	0.900	0.714
Satis2	0.397	0.536	0.626	0.706	0.663	0.715	0.672	0.695	0.910	0.739
Satis3	0.398	0.461	0.634	0.681	0.657	0.691	0.694	0.709	0.918	0.765
Satis4	0.376	0.478	0.630	0.697	0.674	0.688	0.694	0.720	0.887	0.753
Loyal1	0.400	0.477	0.605	0.702	0.626	0.660	0.682	0.674	0.826	0.871
Loyal2	0.284	0.411	0.437	0.533	0.469	0.493	0.513	0.499	0.573	0.741
Loyal3	0.370	0.420	0.466	0.539	0.467	0.539	0.489	0.496	0.586	0.790
Loyal4	0.448	0.414	0.557	0.577	0.514	0.517	0.546	0.521	0.644	0.846

Source: Created by the author.

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